Reinforcers, Rightward Movement, and Prosodic Greed

Abstract: This paper investigates the nature of displacement by looking into a process that shifts DP-internal locative adverbs to the right in Mandar (Austronesian, Indonesian). This process ignores island constraints and displays a range of restrictions that cannot be stated in the terminology of the syntax. I argue that its properties fall into place when it is viewed as an instance of movement in phonology, operating over prosodic structure in response to fundamentally phonological triggers for displacement. The ultimate force of the investigation is to make two points about the architecture of the grammar: first, displacement can occur outside of the syntax, and second, it can be driven by the principle of Greed.

Keywords:

Prosody, Word Minimality, Movement, Greed, Ellipsis

1 Introduction

The investigation of movement holds a special place in linguistic theory for the way in which it sheds light on the broader architecture of the modules where movement occurs. In the syntax, this type of investigation has driven advances in our understanding of phrase structure, locality domains, Case and Agreement, and the organization of the interfaces. This insight unfolds largely from two basic questions: what are the constraints that govern types of movement, and why should they hold as they do?

The goal of this paper is to extend these questions to a specific case of displacement that operates in Mandar, an Austronesian language of Sulawesi. This is a process that targets two locative adverbs which originate inside the DP in the presence of particular demonstratives, on a par with elements like the French *ci* and *là* (1). Following Bernstein 1997, I will refer to these locative elements in this use in Mandar as *reinforcers*.

(1) *The Demonstrative-Reinforcer Construction in Mandar*

Ka'bal	i	de'	[$_{DP}$	**do**	tomauweng	**o**	].
invincible	3ABS	they say		that	grey one	there	

"That old dude there is invincible, they say." Sikki *et al.* 1987, 541

The reinforcers are initially interesting for the following fact: they are obligatory in the presence of certain demonstratives, as if they were lexically selected within the DP, but they are invariably forced to surface at the right edge of a large clause-like domain. The result is that they often seem to undergo steps of movement with the shape in (2).

(2) *Reinforcer Postposing in Mandar*

Na-pittama	i	[$_{DP}$	**do**	setang	____	]	iAli	dionging	**o**.
3ERG-seize	3ABS		that	demon			NAME	yesterday	there

"That demon there seized Ali yesterday."

The real puzzles within this system, however, begin from the following observation: the mechanism that positions the reinforcers shows a locality profile that is very distinct from that which we might reasonably expect from the regular operations of the syntax. To provide an initial glimpse into this system, for instance, the reinforcers appear at the right edges of matrix clauses even when their selecting demonstratives are placed in embedded clauses nested within complex NPs (3)—yielding steps of movement that seem to violate both the Complex NP Constraint and the Right Roof Constraint of Ross 1967.

(3) *The Locality Conditions of Reinforcer Postposing*

Ma'ita a'	[$_{\text{NP}}$ karewa	[$_{\text{CP}}$ mua'	pole	i	**do**	tau	___	]]	di facebook	**o**.
I saw	news	that	come	3ABS	that	guy			on facebook	there

'I saw [$_{\text{NP}}$ news [$_{\text{CP}}$ that that guy there is coming]] on facebook.'

The task of this paper is then to work out a precise theory of this step of movement: one that neatly defines the domains of its application and the position that it targets, describes the constraints on locality that guide it and explains how those might emerge, and ultimately derives its shape from the workings of general systems in the grammar. The investigation that emerges from these desiderata will ultimately lead us deep into the realm of clause-level prosodic organization, and its result, I argue, will be to show that these reinforcers must be carried to their surface positions in the phonology—by an output-optimizing process that draws them for purely phonological reasons toward a position defined in terms of the surface prosodic organization of the clause. This is a process that I will term Reinforcer Postposing, and its shape is sketched in (4).

(4) *The Rule of Reinforcer Postposing*

$\{_{\iota}$ … $\{_{\phi}$ *dem* N ___ $\}$ … REINFORCER $\}$

The broadest consequence of this investigation, then, is to adduce another argument that it is fundamentally possible for the phrasal phonology to relinearize constituents in surface phonological representations in response to purely output-oriented constraints.

In this way, the facts of Mandar contribute to the wider theoretical push to reinterpret displacement as a mechanism that emerges repeatedly across many different modules of the grammar—from the syntax to the morphology (Embick & Noyer, 2001) and just as well to the phonology (McCarthy & Prince, 1993; Halpern, 1995; Hargus & Tuttle, 1997; Chung, 2003; Harizanov, 2014; Aissen, 2017; Bennett *et al.*, 2016; Kusmer, 2020).

When we look further into the phonological shape of this system, in turn, we will uncover a conspiracy in the prosodic phonology that leads to a finer theoretical point: the reinforcers carry within them a phonological tension that demands resolution, and the process of Reinforcer Postposing emerges as a strategy to place them in the singular position where elements of their shape can exist. In other words, it would seem that this process is forced by a phonological type of GREED (Chomsky, 1995b; Bošković, 1995): it is driven by a relatively delicate prosodic property of the moving elements themselves.

In wider perspective, this too is a valuable result. In the current theoretical context, it is standard to assume that syntactic movements are predominantly triggered by the needs of their landing sites—invoking ATTRACT (Chomsky, 2001)—or by the pressure to escape certain positions (yielding PUSH-movement: Diesing 1992; Moro 2000; Chomsky 2013). In the same way, the clearest cases of movement in the phonology have all been argued to emerge from PUSH factors (Halpern, 1995; Bennett *et al.*, 2016; Kusmer, 2020). The case that emerges from our study of Mandar, then, is one which reinforces the case that movement in the phonology—and thus movement in cross-modular perspective—can be driven by the principle of GREED.

The remainder of this paper is organized as follows. Section 2 provides background information on Mandar and Reinforcer Postposing. Section 3 runs our process through a gauntlet of syntactic analyses and Section 4 lays out the goals for our analysis to meet. Section 5 then argues that Reinforcer Postposing must operate in the phonology; Section 6 makes the case for Greed; Section 7 resolves remaining issues and Section 8 concludes.

2 Background

Mandar is an Austronesian language of the South Sulawesi subfamily, spoken by 400,000 people in the Indonesian province of West Sulawesi (Grimes & Grimes, 1987). It is a verb-initial language, and across all contexts it shows a basic word order of V-S-O-D-X (verb > external argument > internal argument > applied argument > adjuncts). The language allows for *pro*-drop and shows no morphological case-marking, but it has a system of agreement: transitive verbs always agree with their external arguments, and every finite clause contains a second-position clitic that spells out absolutive agreement—targeting the internal arguments of transitive verbs and the sole arguments of intransitive verbs. The Mandar verb then shows alternations in transitivity that yield a Western Austronesian "voice system" (Lee, 2008), in which different types of arguments are able to trigger absolutive agreement (Brodkin 2022b; elsewhere see Guilfoyle *et al.* 1992; Aldridge 2004). A regular clause of the language is shown in (5).[1]

(5) *The Shape of a Mandar Clause*

Na-alli-ang	i	[$_S$	iKaco'	]	[$_O$	bunga	]	[$_D$	iCicci'	] dionging.
3ERG-buy-APPL	3ABS		NAME			flower			NAME	yesterday

'Kacho' bought Chichi' flowers yesterday.'

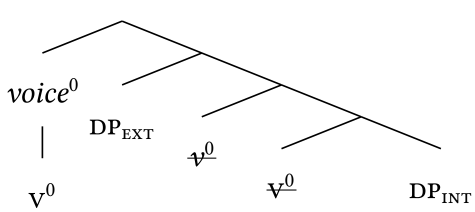
Mandar has been the subject of much descriptive work by the Indonesian Language Office of South Sulawesi, including a grammar (Pelenkahu *et al.*, 1983), a description of adverbs (Sikki *et al.*, 1987), a compilation of traditional poetry (Muthalib & Sangi, 1991), and a conversational handbook (Friberg & Jerniati, 2000). The essential syntax of voice, Case, and agreement in the language has been worked out in Brodkin 2021a,b, 2022a,b. Much of the word-level phonology is described in Jerniati 2005 and in the grammar of Pelenkahu *et al.* 1983; aspects of the phrasal phonology are discussed in Brodkin 2024a,b.

From a comparative perspective, it can also be compared to the other languages of the South Sulawesi subfamily, described in a larger body of English-language work (Friberg, 1991, 1996; Strømme, 1994; Matti, 1994; Valkama, 1995a,b; Jukes, 2006; Lee, 2008; Kaufman, 2008; Laskowske, 2016; Finer, 1997, 1998, 1999; Béjar, 1999).

The system under investigation in this paper has not been described in prior work on the language, though its shape can be seen in data gathered by descriptive sources. The unsourced judgments in this paper have been gathered over five years of work with Jupri Talib, a native speaker from the town of Ugibaru. These judgments have overall been decisive and consistent, and the key patterns that emerged from this work were all reviewed and reconfirmed in the summer of 2022 with Jerniati, the former head of the Indonesian Language Office of South Sulawesi and a native speaker of Mandar. At the same time, many were replicated live and without incident on the Mandar cultural podcast *Pada detik ini*, hosted by Ridwan Alimuddin, an author and cultural authority. It is my understanding that the data below are representative of standard Mandar.

As we launch this investigation, it will be useful to begin from the following fact. Brodkin 2022b argues that the basic word order of Mandar emerges from a process of head-movement that carries the verb above all of the arguments in the extended VP—and more specifically, places it above the head that introduces the external argument (landing in *voice*^{0}, above v^0; on the labels of these heads, Collins 2005; Merchant 2013). The following tree thus illustrates the relevant syntax of a basic transitive clause.

(6) *The Derivation of VSO Order*



2.1 Two Types of Locative Adverbs

Mandar has a class of ordinary locative adverbs whose behavior is relatively routine. This class includes the four elements *indi* "here (immediately by the speaker)", *dini* "here (farther away but still near the speaker)", *dio* "there (farther away from the speaker)", and *diting* "there" (even farther away from the speaker and necessarily out of sight)". These elements are presented in a listed fragment context in (7).

(7) *Disyllabic Locative Adverbs in Mandar*

indi, dini, dio, diting.
here, here, there, there
'here (very close), here (somewhat close), there (somewhat far), there (very far)'

These locative adverbs are distributed in the same way as regular adjuncts to the extended VP in Mandar. In VSO clauses, then, they follow the verb and its arguments (8).

(8) *Disyllabic Adverbs: Follow the VP*

[$_{voiceP}$ [$_{voiceP}$ Ma'-alli a' bau sola barras] **dio** di pasar].
ANTIP-buy 1ABS fish and rice there in market

'I was buying fish and rice there in the market.' Sikki *et al.* 1987, 362

In this same position—at the right edge of the extended VP—there are two other locative elements that can appear. These are the monosyllabic *e* "here" and *o* "there" (9), whose meanings are exactly synonymous with *dini* "here" and *dio* "there," respectively.

(9) *The Monosyllabic Adverbs*

a. Urang i e.
rain 3ABS here
'It's raining here.'

b. Urang toi o.
rain also.3ABS there
'It's also raining THERE.'

The monosyllabic adverbs *e* "here" and *o* "there" are syntactically distinct from the disyllabic adverbs in (12), and it is possible—though never obligatory—for adverbs from the two classes to co-occur. Thus at the right edge of a VP it is possible to adjoin a disyllabic adverb like *diting* "there" and then a semantically-matching monosyllabic adverb like *o*—in the strict and invariant order of DISYLLABIC > MONOSYLLABIC.

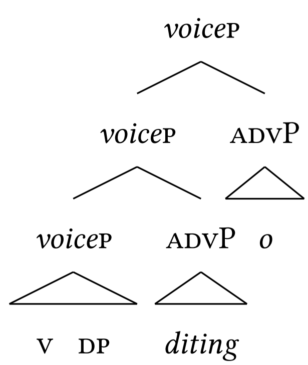
(10) *Co-occurrence: Disyllabic Adverbs + Monosyllabic Adverbs*

Mala	bappa	i	[$_{voiceP}$	na-tarima	akkatta-ta'	**diting**	**o**	].
can	hopefully	3ABS		3ERG-accept	purpose-2GEN	there	there	

'Hopefully they can accept your purpose there.' Friberg & Jerniati 2000, 243

In cases like this, it seems reasonable to imagine that we have the syntax in (11): the disyllabic adverb adjoins to the *voice*P and the monosyllabic adverb adjoins above that.

(11) *Locative Adverbs: Adjoined to the Voice*P



2.2 Demonstratives and Reinforcers

Beyond their use as right-adjuncts to the extended VP, the disyllabic and monosyllabic locative adverbs can also be used in a second syntactic context: within the extended NP (or the DP, for short). This use can be seen in contexts of deixis, where the disyllabic and monosyllabic adverbs can surface after an associated demonstrative and NP—on an exact par with the parallel construction in English (*that cat there*) and French (*ce chat là*).

(12) *Locative Adverbs within the DP*

a. [$_{\text{DP}}$ Do posa o].
that cat there
'That cat there.'

b. [$_{\text{DP}}$ Do posa dio o].
that cat there there
'That cat over there.'

Our investigation begins in this arena from a property of the demonstrative *do*. Mandar is a language that lacks an overt definite article, and in the contexts where nominals refer back to referents already introduced in the discourse, the language requires that they surface with the demonstrative *do* "that." But in this context, the language also requires the appearance of the monosyllabic adverb *o* (13b)—despite the fact that in the context of anaphora, in both Mandar and English, it is semantically anomalous to introduce an "ordinary" locative adverb like *dio* "there" into the DP (13c).

(13) *The Demonstrative Do → The Adverb O*

a. Diang 2 analisa di-timbangngi: 1 sittaksis, 1 ponologi
THERE ARE 2 analysis 1PL.ERG-consider 1 syntactic 1 phonological
'There are 2 analyses we're considering: one syntactic, one phonological.'

b. Alus i [$_{\text{DP}}$ do analisa ponologi o/*____].
elegant 3ABS that analysis phonological there
ROUGHLY: 'The phonological analysis is elegant.'

c. #Alus i [$_{\text{DP}}$ do analisa ponologi dio o].
elegant 3ABS that analysis phonological there there
#'That phonological analysis over there is elegant.'

The deeper generalization beneath this fact is a requirement for co-occurrence: in every context where we see the demonstrative *do*, we invariably also see *o* (with two exceptions discussed in Sections 3 and 7). The same requirement thus emerges in cases deixis: whenever an NP takes the demonstrative *do*, there too we always see *o*.

(14) *Requirements for Cooccurrence*

Na-saka i posa [$_{\text{DP}}$ do balao o/*___].
3ERG-catch 3ABS cat that mouse there
'The cat caught that mouse there (pointing to it).'

This initial restriction, in turn, forms part of a wider system of lexical selection. Beyond the element *do*, Mandar has three more demonstratives: *ndi* "this (extremely close to the speaker)," *de* "this (farther away but still near the speaker)" and *iting* "that (very far away; necessarily out of sight)." And just as the presence of *do* forces the presence of *o*, so too does the presence of either *indi* or *de* necessitate the presence of *e*.

(15) *Two More Selectional Restrictions*

a. [$_{\text{DP}}$ Ndi posa e/*___].
this cat here
'This cat here (which I am currently holding up).'

b. [$_{\text{DP}}$ De posa e/*___].
this cat here
'This cat here (which is somewhere near me).'

The result is that we seem to be facing the network of selectional restrictions in (16): in an idiosyncratic and lexically specific fashion, the demonstratives *ndi, de,* and *do* require the monosyllabic adverbs *e* and *o*. This property sets these three demonstratives apart from *iting*, and it also sets them apart from the English *that*, French *ce*, and many other types of demonstratives that accept adverbial reinforcers but do not require them.

(16) *The Selectional Restrictions*

DISTANCE	DEM	REINFORCER
closest	*ndi*	e
close	*de*	e
far	*do*	o
farthest	*iting*	_

We can understand these restrictions to emerge from the syntactic relationship of l(exical) selection: the demonstratives *ndi*, *de*, and *do* carry syntactic features that force the introduction of the monosyllabic adverbs *e* and *o*. Following Merchant 2019, we can denote this relationship in the following way: the lexical entries of these demonstratives (set inside square brackets) contain a stack of features (set inside angle brackets) that contain the l-selectional features [• *e* •] and [• *o* •].

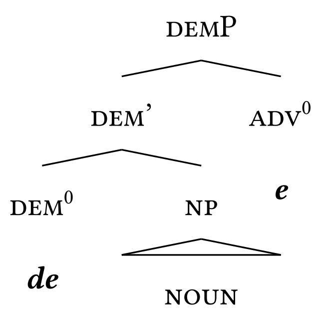
(17) *The Selectional Requirements of the Demonstratives*

a. *ndi*, *de*: $[< \bullet\ e \bullet >]$

b. *do*: $[< \bullet\ o \bullet >]$

This formalization is important for two reasons. The first is terminological: on the view that the monosyllabic locative adverbs are selected by demonstratives in this context, we can fairly call these elements reinforcers—borrowing a term from the literature on parallel constructions in other languages (Bernstein, 1997; Roehrs, 2010). The second is syntactic. When monosyllabic adverbs are selected by demonstratives, we can surmise that the two must sit in the same domain in the syntax—as selection, as a relationship, is characterized by extreme and exacting constraints on locality (Collins & Stabler, 2016). If demonstratives sit somewhere within the dp, then, so too must the *Reinforcers*—that class of monosyllabic adverbs that are selected in this way. I propose that they originate as specifiers of dem^{0}s, which I take to be heads in the dp.[2]

(18) *The Syntax of Selection*



2.3 Separation

We can now advance on the central puzzle of the paper. The reinforcers directly follow their associated noun phrases, or ASSOCIATES, in the fragment contexts that we have seen so far. But there are a great many cases in the language in which the reinforcers are separated from their associates by material that cannot plausibly sit in the DP. For instance, when a reinforcer originates within an associate that is followed by an adjunct to the VP, the reinforcer—though it must be selected by the designated demonstrative inside the associate—must surface after the VP-level adjunct.

(19) *Reinforcers: Separate from their Associates*

Pole boi [$_{DP}$ **do** setang] dionging **o**.
come again.3ABS that demon yesterday there
'That demon there came back again yesterday.'

The same separation is forced whenever an associate is followed by additional arguments in the extended VP. In a VSOD clause, for instance, a reinforcer that is selected by a demonstrative in the S must surface in a position that follows the O and the D (20).

(20) *Reinforcers: the Scale of Separation*

Yari na-bengang i [$_{DP}$ **do** sandro] yima' passikola **o**.
So 3ERG-give 3ABS that shaman talisman schoolkid there
'So that shaman there gave talismans to the schoolkids.'

The same effect emerges whenever an associate is followed by a non-extraposed CP. A reinforcer that is selected in such an associate must take a position that follows the CP (21). We will return to this effect and to the exact syntax of extraposition in Section 5.

(21) *Reinforcers: Separated by Non-Extraposed CPs*

Tapi ma'-ua i [$_{DP}$ **de** sandro] [$_{CP}$ mua' roppong i] **e**.
but ANTIP-say 3ABS this shaman that junk 3ABS here
'But this shaman here says that they're junk.'

The initial force of these facts is to suggest that the reinforcers obey a positional constraint: they must surface at the right edge of a domain that roughly corresponds to the clause. We can summarize this starting generalization along the following lines, with the understanding that we will refine and rethink it as our investigation unfolds.

(22) *Starting Generalization on Distribution*

The reinforcers invariably surface at the right edge of a clause-like domain (one that corresponds to fragments and complete CPs, but not DPs or *voice*Ps).

Beneath this initial generalization, in turn, we can now make a case for movement. There are many cases where it seems plausible to imagine that *e* and *o* are base-generated in the positions where they are pronounced: for instance, in the clauses where they are right-adjoined to the *voice*P (23a) or selected within clause-final associates (23b).

(23) *Base-Generation?*

a.

$[_{voiceP}$ $[_{voiceP}$	Na-bengang	i	sandro	yima'	passikola	]	o	].
	3ERG-give	3ABS	shaman	talisman	schoolkid		there	

'The shaman gave talismans to the schoolkids there.'

b.

$[_{voiceP}$	Na-bengang	a'	$[_{DP}$ **do**	sandro	**o**	]	].
	3ERG-give	1ABS	that	shaman	there		

'That shaman there gave it to me.'

But given the great many cases where the reinforcers must originate within the DP and surface somewhere else, it seems prudent to assume that some process invariably targets *e* and *o* and carries them to their surface positions—applying without regard to their base position and operating even when it has no discernible effect on linear order. Focusing on the cases where this process targets instances of *e* and *o* that are selected in DPs, I will refer to this process as REINFORCER POSTPOSING. Its shape is given in (24).

(24) *Reinforcer Postposing*

[[V [DEM NP ___] ARGUMENTS ADJUNCTS CPS] REINFORCER]

3 Dealing with Separation

At first blush, Reinforcer Postposing seems amenable to several types of analysis in the syntax: we might imagine that the linear positions of the reinforcers emerge from (*i*) a step of movement that attracted the reinforcers to a position in the right periphery, (*ii*) a conspiracy of movements that move everything else in the clause to the left, or (*iii*) base-generation of the reinforcers in their surface positions (ignoring or reframing our conclusions about selection). As we weigh each of these analyses, however, we will gradually come to a case that our operation is completely distinct from all familiar types of movement in the syntax—and instead plays by an idiosyncratic ruleset of its own.

3.1 Rightward movement

The most straightforward analysis of Reinforcer Postposing would be to assume that the adverbs *e* and *o* were systematically attracted to the right edge of the *voice*P, along the lines in (25). I will refer to this approach as the Rightward Syntactic Movement Analysis.

(25) *The Rightward Syntactic Movement Analysis*

[$_{voiceP}$ [$_{voice'}$ V [$_{\text{DP}}$ ASSOCIATE ____] ARGUMENTS ADJUNCTS] REINFORCER]

The Rightward Syntactic Movement Analysis has an immediate and intuitive appeal: it places the reinforcers in the correct position, involves a single step of movement, and looks like a plausible operation that could occur in the syntax. In support of this positive evaluation, we can now add an additional fact: there is an independent process in Mandar that has the shape in (25), and it is a type of focus movement that places its targets at the right edge of the clause. Its effect is shown in the ditransitive clause in (26), where it targets the O *buku ilmu bahasa* "linguistics textbooks"—and delivers the exceptional word order V_$_{\text{O}}$DO. I will refer to this process as Rightward Focus Movement.

(26) *Rightward Focus Movement*

Mam-bengang a' ___ mahasiswa dionging [$_{DP}$ buku ilmu basa].
ANTIP-give 1ABS student yesterday book linguistics

'Yesterday I was giving students LINGUISTICS TEXTBOOKS.'

Developing the parallel a little further, we can go on to note several deeper similarities between Rightward Focus Movement and Reinforcer Postposing. The first of these is a fact of subextraction: Rightward Focus Movement, like Reinforcer Postposing, can subextract specifiers from the DP. We can see this fact with a brief detour into the syntax of possession. There is a class of possessors in Mandar that surface at the right edge of the DP and follow a type of possessor agreement. I propose that these possessors occupy a rightward specifier of D^0 (27a), as they survive the type of NP-ellipsis in (27b).

(27) *Another DP-Internal Specifier*

a. [$_{DP}$ [$_{D'}$ [$_{NP}$ paket] -mu] i'o]
box 2GEN 2SG

'Your box.'

b. Dini i [$_{DP}$ [$_{D'}$ paket-u] yau], tapi pole pai [$_{DP}$ [$_{D'}$ ~~paket-mu~~] i'o].
here 3ABS box-1GEN 1SG but come yet.3ABS box-2GEN 2SG

'My box is already here, but yours is yet to come.'

Rightward Focus Movement can shift these possessors out of the DP (28).

(28) *Rightward Focus Movement can yield Right-Branch Extraction*

U-baca i [$_{DP}$ [$_{D'}$ buku-nna] ___] dionging Suradi Yasil.
1ERG-read 3ABS book-3GEN yesterday NAME

'I read a book yesterday by (the famous Mandar poet) Suradi Yasil.'

This observation sets the stage for an analysis that treats Reinforcer Postposing as a subcase of Rightward Focus Movement. If we accept the view that reinforcers originate in specifier positions in the DP, as proposed in (18), then we can imagine that they might be subextracted from their associates and placed in their surface positions by an obligatory type of Rightward Focus Movement. The shape of this analysis is sketched in (29). I will refer to it as the Unified Rightward Syntactic Movement Analysis below.

(29) *The Unified Rightward Syntactic Movement Analysis*

[$_{\text{FOCP}}$ [$_{\text{VP}}$ V [$_{\text{DP}}$ ASSOCIATE ~~REINFORCER~~] ...] REINFORCER]

We can now make a second observation that would seem to support the Unified Rightward Syntactic Movement Analysis. In the clauses where reinforcers move to the right, it is ungrammatical for other elements to undergo Rightward Focus Movement. The result is that Reinforcer Postposing and Rightward Focus Movement operate in a type of complementary distribution: when a reinforcer moves to the right edge, a focused element cannot (30). I will refer to this pattern as the Complementarity Effect.

(30) *Rightward Focus Movement is Blocked by Overt Reinforcers*

*Mam-bengang i	[$_{\text{DP}}$	**do**	guru]	___	passikola	**poloppeng**	**o**.
ANTIP-give	3ABS	that	teacher		schoolkid	pen	there

'That teacher there was giving schoolchildren PENS'

The most obvious way to understand this pattern of complementarity is as follows: (*i*) these two processes both operate in the syntax and target the same position and (*ii*) they are driven by an attracting head, like c^0, that can only attract a single target. When this head attracts a reinforcer, then, it should be unable to attract a focus (31).

(31) *The Unified Rightward Syntactic Movement Analysis*

[$_{\text{FOCP}}$ [$_{\text{VP}}$ V [$_{\text{DP}}$ DEM NP ~~REINFORCER~~] DP2] REINFORCER]

3.2 Complications

Beneath this initial picture, however, lies a network of effects that suggest that the system at hand is more complex. The first of these lies with the Complementarity Effect itself, which displays a caveat that we will ultimately be forced to derive in Section 7. The caveat is the following: in the clauses where reinforcers are present in the syntax, it is possible for other constituents to undergo Rightward Focus Movement—so long as the reinforcers are not realized overtly. This effect is shown in example (32): in a clause that contains a selecting demonstrative and an argument that undergoes Rightward Focus Movement, the demonstrative is exceptionally not matched by a domain-final reinforcer.

(32) *Rightward Focus Movement forces Reinforcer Deletion*

Mam-bengang	i	[$_{DP}$ **do**	guru	]	___	passikola	**poloppeng**	___.
ANTIP-give	3ABS	that	teacher			schoolkid	pen	

'That teacher there was giving schoolchildren PENS.'

This interaction reveals that the reinforcers obey a second filter on their distribution. In the clauses that contain selecting demonstratives, it seems reasonable to assume that reinforcers should still be generated in derivations like that in (32). To capture their absence on the surface, then, it seems necessary to assume that the reinforcers are not spelled out in derivations where other material undergoes Rightward Focus Movement. At first blush, there are a number of ways to interpret this first pattern—and at present, it will be enough to understand it along the lines in (33). This initial summary will be revised and sharpened as we study similar patterns of deletion in Sections 3.4 and 7.

(33) *Reinforcer Deletion: First Pass*
In derivations where other material is targeted by Rightward Focus Movement, the reinforcers that are generated in the syntax are suppressed at the surface.

With this much in place, we can now turn to the real arena of divergence: the system of locality. Our investigation begins with a brief review of island effects (Ross, 1967)—those restrictions that prevent syntactic movements from escaping coordinate structures, complex NPs, and certain adjuncts, subjects, and complement CPs. The process of Rightward Focus Movement shows a robust and regular sensitivity to island effects, and as a result, it cannot proceed from—for instance—the coordinate structure in (34).

(34) *Rightward Focus Movement respects the Coordinate Structure Constraint*

*U-ita i [$_{\&P}$ sola-na iKaco' na sola-na ___] allo ajuma' iAli.
1ERG-see 3ABS friend-3GEN NAME and friend-3GEN friday NAME
'I saw Kaco's friend and ALI's friend on Friday.'

Reinforcer Postposing ignores this constraint. When reinforcers are generated within DPs that sit in coordinate structures, they continue to move to the right edge.

(35) *Reinforcer Postposing ignores the Coordinate Structure Constraint*

U-ita i [$_{\&P}$ sola-na iKaco' na **do** tau] allo ajuma' **o**.
1ERG-see 3ABS friend-3GEN NAME and that guy friday there
'I saw Kaco's friend and that guy there on Friday.'

We can then replicate this asymmetry with every other type of island in the language. Consider, for instance, the Right Roof Constraint (RRC; Ross 1967): the descriptive observation that embedded clauses form islands for rightward movement. Rightward Focus Movement respects this constraint, and as a result, it cannot shift constituents within embedded clauses to the right of adjuncts in the matrix clause (36).

(36) *The Right Roof Constraint*

*Pepeissang i [$_{CP}$ mua' pole i sola-na ___] di facebook iAli.
Go check if come 3ABS friend-3GEN on facebook NAME

'Go check [$_{CP}$ if a friend ___ is coming] on facebook of Ali's.'

Reinforcer Postposing ignores this constraint too. When reinforcers are generated in DPs within embedded clauses, they can shift across matrix-clause adjuncts to the right.

(37) *Reinforcer Postposing ignores the Right Roof Constraint*

Pepeissang i [$_{CP}$ mua' pole i **do** tau] di facebook **o**.
Go check if come 3ABS that guy on facebook there

'Go check [$_{CP}$ if that guy there is coming] on facebook.'

Tightening the screws further, we can observe that this asymmetry persists even when we embed these islands in other types of islands. The simplest illustration involves the Complex NP Constraint (CNPC; Ross 1967), which prohibits movement out of clauses that are embedded within NPs. This constraint holds in its usual way over Rightward Focus Movement, which is absolutely unable to escape embedded clauses that are embedded within NPs (a step which would violate both the RRC and the CNPC).

(38) *The Complex NP Constraint*

*Ma'ita a' [$_{NP}$ karewa [$_{CP}$ mua' pole i sola-na ___]] di FB iAli.
I saw news that come 3ABS friend-3GEN on FB NAME

'I saw [$_{NP}$ news [$_{CP}$ that a friend ___ is coming]] on facebook of Ali's.'

But Reinforcer Postposing, once again, ignores this constraint entirely. When a reinforcer is generated in an embedded clause that sits within a complex NP, it escapes that clause and that complex NP to surface after the final adjunct in the matrix clause.

(39) *Reinforcer Postposing ignores the Complex NP Constraint*

Ma'ita a' [$_{DP}$ karewa [$_{CP}$ mua' pole i **do** tau]] di FB **o**.
I saw news that come 3ABS that guy on FB there

'I saw [$_{DP}$ news [$_{CP}$ that that guy there is coming]] on facebook.'

3.3 Idiosyncratic Islands

The collective force of these facts is to suggest that there is a meaningful difference between Rightward Focus Movement and Reinforcer Postposing: Rightward Focus Movement obeys the canonical range of island constraints and Reinforcer Postposing does not. To this picture we can now add an additional fact: Reinforcer Postposing seems to respect a second and idiosyncratic class of island constraints of its own.

To build a case for this point, it will be useful to consider a second type of analysis for Reinforcer Postposing: one that derives the domain-final position of the Reinforcers from a conspiracy of movement to the left. Such an analysis might unfold in two ways: (*i*) we might imagine that the reinforcers were systematically stranded in a derivation that required everything else to move to the left, after Kayne 1994 (40a), or alternatively, (*ii*) we might imagine that the reinforcers were forced to move to the left periphery and the rest of the clause required to undergo remnant movement to a higher position (40b). I will refer to these two possibilities as the Leftward Syntactic Movement Alternatives.

(40) *Leftward Syntactic Movement Alternatives*

a. [$_{FP}$ V DP ADJUNCT [$_{VP}$ ~~V~~ [$_{DP}$ ~~DEM~~ ~~NP~~ REINFORCER] ~~ADJUNCT~~]]

b. [$_{FP2}$ V DP ADJUNCT [$_{FP1}$ REINFORCER [$_{VP}$ ~~V~~ [$_{DP}$ ~~DEM~~ ~~NP~~ ___] ~~ADJUNCT~~]]]

These analyses make a clear and important prediction: when the associate of a reinforcer is moved left, the reinforcer should remain in its domain-final position. The following diagrams illustrate the logic of the prediction: the stranding analysis in (41a) and the remnant-movement analysis in (41b) should allow associates to move left alone.

(41) *Leftward Syntactic Alternatives: Predictions on Topicalization*

a. [$_{FP}$ DP [$_{FP}$ V ~~DP~~ [$_{VP}$ ~~V~~ [$_{DP}$ ~~DEM~~ ~~NP~~ REINFORCER]]]]

b. [$_{FP3}$ DP [$_{FP2}$ V DP [$_{FP1}$ REINFORCER [$_{FP1}$ [$_{VP}$ ~~V~~ [$_{DP}$ ~~DEM~~ ~~NP~~ ___]]]]]]

In many cases, this prediction is correct: when an associate moves left, its reinforcer remains in its domain-final position. The following example illustrate with a process of focus-fronting, which places its targets in the left periphery (42a) (Brodkin, 2020). When this process targets an associate, its reinforcer continues to remain domain-final (42b).

(42) *Focus-Fronting: "Strands" Reinforcers*

a.

[$_{\text{FocP}}$	**iAli**	[$_{\text{TP}}$	melo'	u-peroa	___	]	].
	NAME		want	1ERG-invite			

'ALI I want to invite.'

b.

[$_{\text{FocP}}$	**Do**	Ali	[$_{\text{TP}}$	melo'	u-peroa	___	**o**	]	].
	that	NAME		want	1ERG-invite		there		

'THAT ALI THERE I want to invite.'

But there are also cases where this prediction is not correct. The first of these emerges around a process of topicalization, whose essential shape is shown in (43a). This process places its targets in the left periphery, and it is very clear that it involves movement, because it obeys all the usual island constraints. For instance, it cannot proceed from adjunct CPs (43b) (obeying the Adjunct Island Constraint of Ross 1967).

(43) *A Second Step of Movement: Topicalization*

a.

[$_{\text{TopP}}$	**iAli**,	[$_{\text{TP}}$	u-bengang	i	doi'	___	]].
	NAME,		1ERG-give	3ABS	money		

'Ali, I gave some money to.'

b.

[$_{\text{TopP}}$	***iAli**,	[$_{\text{TP}}$	sannang	a'	[$_{\text{CP}}$	mua'	pole	i	___	]	].
	NAME		happy	1ABS		if	come	3ABS			

INTENDED: 'Ali, I'd be happy if ___ came.'

Unlike focus-fronting, topicalization has an impact on the positions of the reinforcers. When an associate is topicalized, its reinforcer is forced to follow it up—and descriptively speaking, to surface at the right edge of the topicalized argument.

(44) *Topicalization cannot strand Reinforcers*

[$_{\text{TopP}}$ **Do** sanaeke **o**, [$_{\text{TP}}$ biasa i m-angino ___]].
that kid there usually 3ABS ANTIP-play

'That kid there$_{\text{TOP}}$ is usually playing'

The same pattern emerges around a process of CP-preposing. In the default case, the edges of CPs are freely crossed by Reinforcer Postposing—and as a result, reinforcers that originate in matrix clauses can freely cross over adjunct CPs that follow (45a). But Mandar allows adjunct CPs to be preposed to a position that falls before the matrix clause, and in that context, they become islands: when a reinforcer is generated in a preposed CP of that type, it is trapped at the right edge and cannot move farther (45b).

(45) *CP-Preposing creates Islands for Postposing*

a. Na-mario i [$_{\text{DP}}$ **do** guru] [$_{\text{CP}}$ mua' pole o mie'] **o**.
will-happy 3ABS that teacher if come 2ABS 2PL here
'That teacher there will be happy if you guys come.'

b. [$_{\text{CP}}$ Mua' pole i [$_{\text{DP}}$ **do** mahasiswa] **o**], na-mario a' ___.
if come 3ABS that student there will-happy 1ABS
'If those students there come, I'll be happy.'

These observations raise technical and non-trivial challenges for both of the Leftward Syntactic Movement Alternatives in (40).[3] Their real importance, however, lies in the fact that they reveal independent barriers to the process of Reinforcer Postposing—barriers which appear not around syntactic islands but at the right edges of topics and preposed CPs. Put differently, the facts of topicalization and CP-preposing show that Reinforcer Postposing must operate within absolute locality domains of its own—and must be sensitive to "islands" of another type. We will return to these facts, and to the precise identity of the domain of Reinforcer Postposing, in Section 5.

3.4 Relative Locality

We can now introduce a parallel deviation within the system of relative locality. Syntactic movements canonically obey the constraint in (46): they must obey relativized minimality (Chomsky, 1973; Rizzi, 1990). The typical way to understand this constraint is in terms of c-command: movement triggered by a head F^0 cannot target a goal β that is c-commanded by another goal α beneath F^0 that is identical in the relevant respects.

(46) *A Syntactic Constraint on Relative Locality*

$[_{\text{FP}}$ ___ $[_{\text{GP}}\ \alpha \ldots [_{\text{HP}}\ \beta \ldots]\ ]\ ]$

✗

If Reinforcer Postposing played by the normal rules of the syntax, then, we would expect to see the pattern of relative locality in (47): in a clause that contained two reinforcers, only the syntactically higher of the two should move to the right edge.

(47) *Reinforcer Postposing: Relative Locality?*

$[_{\text{FP}}\ [_{\text{GP}}\ \alpha \ldots [_{\text{HP}}\ \beta \ldots]\ ]$ ___ $]$

✗

This prediction can be tested in clauses that contain two selecting demonstratives. In clauses of this type, it is ungrammatical for two overt reinforcers to surface at the right edge in any order (48a). Turning back for an instant to the case of adjunction, we can note that the reinforcers also cannot co-occur with their adjoined counterparts (48b).

(48) *One Domain, One Reinforcer*

a. *Bemme i [$_{\text{DP}}$ **do** sanaeke] naung [$_{\text{DP}}$ **ndi** passauang] **o** **e**!
fall 3ABS that kid down this well there here
Intended: 'That kid there fell down this well here!'

b. *U-ita i [$_{\text{DP}}$ **do** tau] **o** e.
1ERG-see 3ABS that guy there here
Intended: 'I saw that guy there here.'

Example (49) shows what happens in clauses that host two selecting demonstratives: the underlyingly rightmost reinforcer surfaces at the right edge and the other deletes.

(49) *Two Reinforcers: One Deletes*

Bemme	i	[$_{DP}$ **do**	sanaeke]	naung	[$_{DP}$ **ndi**	passauang]	___	**e**!
fall	3ABS	that	kid	down	this	well		here

'That kid (there) fell down this well here!'

This pattern reflects the same effect that we noted in our study of Rightward Focus Movement: when two elements compete to surface at the right edge of one domain, a reinforcer that loses must be suppressed. We can be certain that the losing reinforcers are underlying present in this context, however, because they resurface when their associates are moved to separate domains of placement—for instance, by topicalization.

(50) *Topicalization: Rescues Losing Reinforceers*

a.

[$_{DP}$ **Do**	sanaeke]	**o**,	bemme	i	naung	[$_{DP}$ **ndi**	passauang]	**e**!
that	kid	there	fall	3ABS	down	this	well	here

'That kid there$_{TOP}$ fell down this well here!'

b.

[$_{DP}$ **Do**	guru]	**o**,	mam-bengang	i	___	passikola	**poloppeng**.
that	teacher	there	ANTIP-give	3ABS		schoolkid	pen

'That teacher there$_{TOP}$ was giving schoolchildren PENS.'

The result is that we seem to be dealing with a Sour-Grapes Effect, of the kind familiar in phonology (Padgett, 1995; Jardine, 2016): the reinforcers either get what they want—they move to the right edge of their domain—or they delete. We can thus shapen our initial characterization of Reinforcer Deletion in (33) to the revised generalization in (51). We will return to this generalization once again in Section 7.2, and there we will demonstrate that it cannot be understood as a regular process of segmental deletion.

(51) *Reinforcer Deletion: Second Round*
If a reinforcer cannot be realized at the right edge of its domain, it is suppressed.

What is important at present in this system is the following fact: in the cases where two reinforcers compete to move to the right edge of a single domain, the factor that decides which reinforcer will win is not c-command. Instead, it is linear order: the reinforcer that surfaces at the right edge is the one whose associate is closest to the right. In example (49), this means that a reinforcer that originates in the clause-final PP *naung ndi passauang* 'down this well' beats a reinforcer that originates in the absolute subject *do sanaeke* 'that child.' This linear competition is sketched in example (52).

(52) *Reinforcer Postposing: Relative Locality via Linear Distance*

[$_{\text{FP}}$ [$_{\text{GP}}$ α ...[$_{\text{HP}}$ β ...]] ___]

✗

The primacy of linear order can be seen clearly in contexts of recursive embedding. It is possible to embed a constituent that contains a selected reinforcer within another constituent that contains a selected reinforcer of its own. In this configuration, the facts of c-command and linear order will conflict: the unembedded reinforcer will be generated farther to the left, and the embedded reinforcer will be generated farther to the right. In this context, the rightmost reinforcer will still win out. This effect is shown in example (50) with the complex NP *de potona solana do tau* 'this photo of the friend of that guy': there, it is the reinforcer associated with *do tau* 'that guy' that survives (53).

(53) *Reinforcer Competition Ignores c-command*

✗

U-olo'	i	[$_{\text{DP}}$ **de**	poto-na	[$_{\text{DP}}$ sola-na	[$_{\text{DP}}$ **do**	tau	]]]	___	**o**.
1ERG-like	3ABS	this	photo-3GEN	friend-3GEN	that	guy			there

'I like this photo (here) of the friend of that guy there.'

This linear competition interacts with syntactic movement in a transparently surface-oriented way. To illustrate, Mandar has a process of rightward scrambling that right-adjoins DPs to the TP. The following examples show how this process affects the competition between reinforcers. In a VSO clause where both the S and the O contain demonstratives, the reinforcer generated in the O will always win out (54a). But when the S in such a clause is scrambled to the right, it is the reinforcer that is generated in the S—now linearly rightmost before postposing occurs— that surfaces at the edge (54b).

(54) *Reinforcer Competition: Transparently Sensitive to Movement*

a. [$_{voice}$ Na-saka i [$_{DP}$ **do** posa] [$_{DP}$ **de** balao]] digena' ___ **e**!
3ERG-catch 3ABS that cat this mouse earlier here
'That cat (there) caught this mouse here earlier.'

b. [$_{voice}$ Na-saka i __ [$_{DP}$ **de** balao]] digena' [$_{DP}$ **do** posa] ___ **o**!
3ERG-catch 3ABS this mouse earlier that cat there
'That cat there caught this mouse (here) earlier.'

These facts raise a steep challenge to an alternative analysis that would base-generate the reinforcers in the right periphery and link them to demonstratives via AGREE, along the lines in (55). This type of analysis has been ill-suited for this system from the outset, as the reinforcers have referential content and can appear on their own. But we can now see that such an account would also necessitate a deeply aberrant form of AGREE: one that could (*i*) see into complex DPs (53), finite CPs (37), and finite CPs embedded in DPs (39), (*ii*) ignore C-command and make reference to pure facts of linear order, (*iii*) and surface within a X^0 that appeared exclusively in matrix clauses, topics (44), preposed adjunct CPs (45), and fragments (15). It is technically possible to formulate a proposal along these lines, but it is not at all clear how such a move could lead us to any deeper insight into the nature of Reinforcer Postposing or the nature of AGREE.

(55) *Rejected: an Agree Analysis*

[$_{FP}$ [$_{VP}$ V [$_{DP}$ DEM NP [$_{DP}$ DEM NP [$_{DP}$ DEM NP]]]] $F^0$$_{\text{“REINFORCER”}}$]

Turning back to the domain of relative locality, then, we arrive at a final result: once again, Reinforcer Postposing ignores the regular rules of the syntax and plays by constraints that are formulated in fundamentally different terms. It is the task of the following sections to understand how a process of this shape might come to exist.

4 An Interim Summary

Taking stock of the picture so far, we have seen that Mandar has a pair of monosyllabic locative adverbs that are lexically selected by certain demonstratives—as reinforcers—but are routinely separated from their associates by a process that places them at the right edges of particular domains. This process operate in complementary distribution with Rightward Focus Movement but obeys idiosyncratic rules of its own: it freely escapes islands, violating the CSC, RRC, and CNPC, and it ignores C-command in the calculus of relative locality. In the same vein, it finds "islands" of its own in topics and preposed CPs and it obeys a form of relative locality that is sensitive to linear order alone.

Against this backdrop, the task before us is to construct an analysis which explains the basic properties of Reinforcer Postposing and integrates these into a broader theory of movement. At the lowest analytical level, this analysis must attempt to:

1. Provide a unified and exhaustive characterization of the domains of placement, which separates the constituents that host reinforcers from those which do not,
2. Explain how this movement can skirt all syntactic constraints on locality, and
3. Explain why this movement obeys the idiosyncratic constraints that it does.

At a higher level, a successful analysis should do the following:

4. Connect the motivation for postposing to independent properties of the grammar,
5. Capture the Sour-Grapes effect, explaining why reinforcers delete if non-final, &
6. Explain the complementarity of Reinforcer Postposing & Rightward F-Movement.

And finally, the analysis must also reach to:

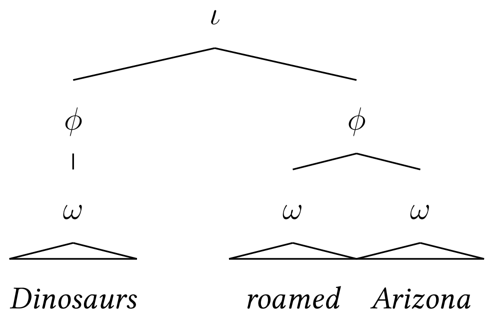
7. Situate itself within a general theory of displacement in the grammar, and
8. Bring its results to bear on the theory of the distribution and motivation of Move.

The remainder of this paper is an attempt to meet the desiderata above. To this end, we will pursue a simple line of attack. Setting the syntax aside, we will investigate Reinforcer Postposing through the lens of the phonology. Our ultimate goal in this connection will be to work out an analysis of that process that operates in fully phonological terms: one which mobilizes the phrasal phonology to explain the constraints, the domains, and the motivation for Reinforcer Postposing. We will then link the results that coalesce in this investigation to a general theory of movement.

5 Phonological Displacement

We begin this second arc of our investigation with the theory of the Prosodic Hierarchy: that framework which studies the ways in which phonological strings are parsed into hierarchical constituent structures in the surface phonology (Selkirk, 1984, 1986, 2009; Nespor & Vogel, 1986; Itô & Mester, 2007, 2012, 2013, 2019). The constituent structure that is relevant for our purposes is of a kind with the hierarchical organization that is known at the metrical level, where segments are parsed into syllables and syllables are parsed into feet (Liberman & Prince, 1977). It is one that is built from three types of recursible and weakly-layered phonological constituents that sit above the metrical level: the prosodic word (ω), the phonological phrase (ϕ), and the intonational phrase (ι) (Itô & Mester, 2007; Selkirk, 2009). The rough shape and distribution of these categories are sketched below in the English example in (56), which shows the parse of the string *Dinosaurs roamed Arizona.* This string is parsed into three ωs ("Dinosaurs," "roamed," and "Arizona"), those words are parsed into two ϕs ("Dinosaurs" and "roamed Arizona"), and those phrases are then parsed into a single higher ι.

(56) *Phonological Organization above the Word*



The essential task of this section is to establish a Mandar-internal strategy to detect a single prosodic constituent of this type—the intonational phrase (ι)—and to show that the ι defines the domain of Reinforcer Postposing. In the following section, then, we will develop a theory that forces the reinforcers to move to the right edge of this prosodic constituent—operating in the phonology and occurring for reasons that will ultimately be grounded in the phonology itself. In keeping with the tradition of research on prosodic phonology (Nespor & Vogel, 1986), we will restrict our attention throughout this investigation to the type of prosody that emerges most readily when speakers are asked to introspect on the most natural pronunciation of particular strings. Following a methodology established in previous work with the same speakers of Mandar (Brodkin, 2024a), we will thus establish the distribution of ιs by leveraging the introspective judgments that speakers can provide on the optimal application of processes in the phrasal phonology. In doing so, we will abstract away from the many factors that derail default patterns of prosodic phrasing in contexts of natural production (Snedeker & Trueswell, 2003; Aylett & Turk, 2004; Watson *et al.*, 2006; Féry & Ishihara, 2016) to focus on the phonological constituency that emerges when speakers are asked to pronounce well-planned strings at a regular speech rate under broad focus.

5.1 A Correlation with the Phonology

Brodkin 2024a documents a number of phonological restrictions that make it possible to identify the edges of specific phonological constituents along the prosodic hierarchy in Mandar, and one restriction from this class is relevant to our present work. In Mandar, Pater 1999 observes that in coda positions the segment /ŋ/ must assimilate in place to all following segments and denasalize before all non-nasal segments except */b d d͡ʒ g/*. Mills 1975, the source of Pater's data, notes that this process applies without exception in the Mandar ω, and as a result, notes that prefix-final instances of /ŋ/ must assimilate to the onsets of following roots (57a). But as we can see from example (57b), the same process operates in the same obligatory fashion across the boundaries of prosodic words.

(57) *Nasal Assimilation in Mandar*

a.

ma**m**-bat͡ʃa,	ma**p**-poloŋ,	mi**n**-daiʔ,	mi**t**-tama,	sa**ŋ**-gallas,	sa**k**-kat͡ʃa
maŋ-bat͡ʃa	ma**ŋ**-poloŋ	mi**ŋ**-daiʔ	mi**ŋ**-tama	sa**ŋ**-gallas	sa**ŋ**-kat͡ʃa
ANTIP-read	ANTIP-cut	VBLZ-up	VBLZ-in	ONE-cup	ONE-glass

'Read, cut, go up, go in, one cup, one glass'

b.

dia**k**	kapa**t**	teɹoŋ.
diaŋ	kapaŋ	tedoŋ
THERE ARE	maybe	water buffalo

'Maybe there are water buffalo.'

The relevance of this process emerges from two generalizations on its distribution. First, Nasal Assimilation always applies between all the words that fall within a string that forms a domain for Reinforcer Postposing. The essential generalization is shown in the following pair of clauses, which contain the associate *do sappaŋ* "that sampan (a type of boat)." In example (58a), this argument appears in its regular postverbal position, and in that context, (*i*) the final ŋ assimilates to the initial segment of a following adjunct and (*ii*) its reinforcer crosses over the adjunct to move to the right edge. Example (58b) then shows a similar pattern in a clause where the same associate is focus-fronted: there, (*i*)

Nasal Assimilation occurs between the associate, the following verb, and the following adjunct and (*ii*) the reinforcer crosses over the verb and the adjunct as it moves right.

(58) *Nasal Assimilation: Occurs within Domains for Postposing*

a. jari i tallan **do** sappa**t** ton djoloʔ **o**.
jari i tallaŋ do sappaŋ tauŋ dioloʔ o
DID 3ABS sink that sampan year last there
'That sampan there did sink last year.'

b. **do** sappa**t** talla**t** ___ ton djoloʔ **o**.
do sappaŋ tallaŋ tauŋ dioloʔ o
that sampan sink year last there
'THAT SAMPAN THERE sank last year.'

Second, this process is blocked at the junctures that halt Reinforcer Postposing. We can see this fact with a brief return to the domains that form islands for Reinforcer Postposing: topics and preposed adjunct CPs. In Section 3.3, we saw that when the reinforcers are generated within topics and preposed adjunct CPs, they are invariably trapped inside of them and cannot move across their right edges. We can now observe that the same right edges *also* block our process of Nasal Assimilation—and thus in topics and preposed adjunct CPs, final coda /ŋ/ is forced to remain intact.

(59) *Nasal Assimilation: Blocked at the Edges that Block Postposing*

a. itit tau**ŋ**, **p**ole i iramas sola wenena .
itiŋ tauŋ pole i iramaŋ sola baine-na
that year come 3ABS NAME with wife-3GEN
'That year, Ramang came with his wife."

b. mwaʔ so i irama**ŋ**, **s**o toaʔ.
muaʔ sau i iramaŋ sau to-aʔ
if go 3ABS NAME go also-1ABS
'If Ramang goes, I'll go too."

The result is that there is a straightforward and transparent correlation between the distribution of Nasal Assimilation and the distribution of Reinforcer Postposing.

The following diagram schematizes the essential result: representing the application of Nasal assimilation with the checkmark ✓, we can say that this process applies between all of the words that fall within the strings that form domains for Reinforcer Postposing and then fails to apply across the right edges of those same domains.

(60) Nasal Assimilation + Reinforcer Postposing: A Single Domain

{ … ✓ … ____ … ✓ … ✓ … *e/o* } ✗ { … ✓ … ✓ … }

In light of this conclusion, it seems natural to conclude that Nasal Assimilation and Reinforcer Postposing are constrained by the same type of constituent structure—and more specifically, to conclude that these processes are keyed to the very same domain. From the perspective of the syntax, we have already noted that this domain corresponds to a motley bunch of elements: fragments, matrix CPs, topics, and preposed adjunct CPs. But given that this domain must also be relevant to the process of Nasal Assimilation—an irreducibly phonological operation—we should begin to wonder whether the relevant type of constituent might be better defined in the terminology of the phonology.

5.2 The Prosodic Generalization

It is against this backdrop that we return to the theory of the Prosodic Hierarchy. Brodkin 2024a argues that there is a straightforward way to state the distribution of Nasal Assimilation in Mandar in surface-oriented terms: coda /ŋ/ assimilates to the following segment, across ω- and ϕ-edges, unless it is final in the intonational phrase (ι). The ensuing characterization of the distribution of our process is shown in diagram (61).

(61) Nasal Assimilation: Blocked at the Edge of the Intonational Phrase

$\{_\iota$ … ✓ … ✓ … ✓ … } ✗ $\{_\iota$ … ✓ … ✓ … ✓ … }

The case for this claim—that Nasal Assimilation operates across ω- and ϕ-edges and is blocked exactly at the right edge of the ι—emerges from the following chain of logic. (*i*) As Nasal Assimilation is a phonological process, its distribution must be defined in phonological terms (the principle of SYNTAX-FREE PHONOLOGY: Nespor & Vogel 1986), (*ii*) Nasal Assimilation is not blocked at the edges of constituents that sit lower on the prosodic hierarchy, like the ω, the ϕ, or the maximal ϕ (the highest instance of the ϕ in a recursive-ϕ structure), which can be seen with other tests in Mandar (Brodkin, 2024a), (*iii*) Nasal Assimilation operates within a domain that is larger than the maximal ϕ, and the only possible candidate for such a larger domain is the ι, on the universal theory of prosodic categories proposed by Itô & Mester 2007 and adopted by Selkirk 2009, 2011. On theory-internal grounds, then, we can surmise that this process is bounded by the ι.

We can then supplement this initial case with a second line of argumentation: (*iv*) In an impressionistic but immediately intuitive way, the strings that form domains for Nasal Assimilation in Mandar all sound exactly like the prosodic constituents that are identified as ιs in English: they are followed by short pauses that sound exactly like "comma intonation" (the pause that is identified as break index 4 in English ToBI). On empirical grounds, then, we can be certain that Nasal Assimilation is blocked at the edge of a prosodic constituent that can be detected from independent phonetic effects. (*v*) The distribution of this comma intonation seems to be the same in Mandar, English, and Indonesian, in that it falls in all of these languages after topics and preposed adjunct CPs. In the framework of Match Theory (Selkirk, 2009, 2011), then, it seems natural to imagine that—for reasons yet unknown—these constituents are universally mapped to ιs by constraints of the family SYNTAX-TO-PROSODY MATCH. If this line of reasoning is on the right track, then, we arrive at a relatively secure case for the claim in (62).

(62) The Distribution of Nasal Assimilation

Nasal Assimilation is blocked systematically and exclusively by the edges of the ι.

The natural consequence of this conclusion is to raise the further possibility in (63): could it be that the same constituent defines the domain of Reinforcer Postposing?

(63) The Domain of Reinforcer Postposing: Phonological Hypothesis

Reinforcer Postposing targets the right edge of the ι.

We can begin to test this hypothesis with a brief pivot to the system of extraposition. Mandar has a process of extraposition that can place many types of constituents—places adverbs, referential arguments, and all types of non-root CPs—in a position that is set off from the matrix clause by comma intonation (on a par with the prosodic consequences of extraposition in many other languages: Antinucci & Cinque 1977; Royer 2022). In this context, the distribution of Nasal Assimilation shows that the extraposed constituents form their own ιs—a parse that is indicated in the top lines of the examples below.

(64) *Extraposition in Mandar*

a. {$_\iota$ wita i iramaŋ ___ } {$_\iota$ **t**on djolo }
u-ita i iramaŋ taŋ diolo
1ERG-see 3ABS NAME year last
'I saw Ramang, last year.'

b. {$_\iota$ nopelambi i iramaŋ ___ } {$_\iota$ **t**enneʔ moŋeʔ i }
na-u-pelambiʔi i iramaŋ tenneʔ moŋeʔ i
will-1ERG-visit 3ABS NAME in case sick 3ABS
'I'll visit Ramang, in case he's sick.'

c. {$_\iota$ napelambi i iramaŋ ___ } {$_\iota$ **k**indoʔna }
na-pelambiʔi i iramaŋ kindoʔ-na
3ERG-visit 3ABS NAME mom-3GEN
'She visited Ramang, his mom.'

This footwork sets up the following prediction. If Reinforcer Postposing places its targets at the right edge of the ι, then we should expect that the positions of reinforcers will change when the boundaries of the ι are manipulated by extraposition. And this prediction is correct: the reinforcers are forced to precede all extraposed material.

(65) *Reinforcer Postposing is sensitive to Extraposition*

a. $\{_\iota$ wita i ɹo rama**ŋ** **o** } $\{_\iota$ **t**on djolo }.
u-ita i do ramaŋ o taŋ diolo
1ERG-see 3ABS that NAME there year last

'I saw that Ramang there, last year.'

b. $\{_\iota$ nopelambi i ɹo rama**ŋ** **o** } $\{_\iota$ **t**enneʔ moɲeʔ i }
na-u-pelambiʔi i do ramaŋ o tenneʔ moɲeʔ i
will-1ERG-visit 3ABS that NAME there in case sick 3ABS

'I'll visit that Ramang there, in case he's sick.'

c. $\{_\iota$ napelambi i ɹo rama**ŋ** **o** } $\{_\iota$ **k**indoʔna }
na-pelambiʔi i do ramaŋ o kindoʔ-na
3ERG-visit 3ABS that NAME there mom-3GEN

'She visited that Ramang there, his mom.'

The same effect can be seen in the behavior of complement CPs. In Mandar, many bridge verbs require their complement CPs to undergo extraposition, and we can now read this fact directly from the distribution of Nasal Assimilation: this process shows, for instance, that the complement CPs of *sanga* "think," must form their own ιs (66a). In this context, the reinforcers behave in their expected way: when they originate in the matrix clause, they shift to the edge of its corresponding ι to fall before the extraposed CP (66b).

(66) *Extraposition of Complement CPs*

a. $\{_\iota$ nasaŋa irama**ŋ** } $\{_\iota$ **t**allaŋ i }.
na-saŋa iramaŋ tallaŋ i
3ERG-think NAME sink 3ABS

'Ramang thinks, that it sank.'

b. $\{_\iota$ nasaŋa **ɹo** ramaŋ **o** } $\{_\iota$ tallaŋ i }.
na-saŋa do ramaŋ o tallaŋ i
3ERG-think that NAME there sink 3ABS

'That Ramang there thinks, that it sank.'

Despite this fact, there is a context in which CP-extraposition is visibly blocked: when an embedded CP launches WH-movement, it is never extraposed and invariably forms a single ι with the matrix clause. This fact can be seen, once again, from the distribution of Nasal Assimilation—which exceptionally begins to apply across the edge of the embedded CP in this context (67a). This shift, in turn, correlates with an explicit change in the distribution of Reinforcer Postposing: when extraposition is blocked in this way, reinforcers that originate in the matrix clause cross over embedded CPs (67b).

(67) *Reinforcers Change Positions when Extraposition is Blocked*

a. | $\{_\iota$ | a | nasaŋa | iramat | tallaŋ | ___ | }? |
|---|---|---|---|---|---|---|
| | a | na-saŋa | iramaŋ | tallaŋ | | |
| | what | 3ERG-think | NAME | sink | | |

‘What does Ramang think sank?’

b. | $\{_\iota$ | a | nasaŋa | **ɹo** | ramat | tallaŋ | ___ | **o** | }? |
|---|---|---|---|---|---|---|---|---|
| | a | na-saŋa | do | ramaŋ | tallaŋ | | o | |
| | what | 3ERG-think | that | NAME | sink | | there | |

‘What does that Ramang there think sank?’

The picture that emerges from this initial investigation, then, is one on which there is a transparent phonological generalization that hangs above Reinforcer Postposing: this process interacts with topicalization, CP-preposing, extraposition, and all other types of movement in a manner that correlates directly with the distribution of ιs. The result is that we can summarize the shape of this process as follows: Reinforcer Postposing carries its targets to the right edge of the smallest ι that contains the associate and the base position of the associate: a topic, a preposed CP, or a matrix clause.

(68) *The Phonological Shape of Reinforcer Postposing*

$\{_\iota$ … } $\{_\iota$ … $\{_\phi$ *dem* N ___ } … REINFORCER } $\{_\iota$ … }

5.3 Movement in the Phonology

It is at this point in our investigation that we can take a more aggressive empirical step. Research in the paradigm of Prosodic Hierarchy Theory has long recognized that there is a relatively close relationship between the suprametrical constituent structure of the phonology and the hierarchical structure of the syntax beneath it: beneath the level of the ι, for instance, ωs typically correspond to lexical x^0s and ϕs typically correspond to XPs (Truckenbrodt, 1995, 1999; Selkirk, 2009, 2011). As a result, it is at least conceivable that our new prosodic generalization in (67)—that Reinforcer Postposing systematically targets the right edge of the ι—could actually reflect a fact of syntax: perhaps Reinforcer Postposing is a syntactic operation that systematically targets a position—or some set of positions— that will eventually fall at the right edge of an ι when the syntactic derivation is passed off to the phonology. We can summarize this idea along the following lines.

(69) *The Final Syntactic Attempt*

Reinforcer Postposing is a syntactic operation that places its targets in positions that will invariably fall at the right edges of ιs when prosodic structure is built.

The task of this subsection is to consider and reject the syntactic hypothesis in (69). Our strategy will be straightforward. One of the chief discoveries of research on the Prosodic Hierarchy is that suprametrical prosodic domains are not always isomorphic to the syntactic constituents beneath them. The constituent structure of the phonology, rather, can deviate from the constituency of the syntax in response to pressures native to the phonology and the syntax-prosody interface—such as constraints on balance and accentual clash (Kubozono, 1989; Selkirk & Elordieta, 2010) and constraints that govern the prosodification of particular relationships in the syntax (Kalivoda, 2018). The result is that there are cases where prosodic junctures—like the edge of the intonational phrase—will surface in positions that cannot be predicted in any way from the pure facts of syntax. This step opens up a path to a new type of argument: if we can show that Reinforcer Postposing places its targets at the edges of intonational phrases whose boundaries must be fixed in the phonology, we can demonstrate that it must operate over a constituent structure that is fundamentally oriented toward the surface.

It is against this backdrop that we turn to the particular prosody of appositives. Mandar, like English, allows appositive phrases to be adjoined to nominal arguments, and their introduction triggers a change in prosodic phrasing that is both phonologically visible and intuitively clear. Whenever an appositive is adjoined to an argument, the appositive forms its own ι and the surrounding clause is split into multiple ιs (70b).

(70) *The Prosodic Effect of Appositive Insertion*

a.

{$_\iota$ kessaŋa	ĩ	irama**k**	**k**orupsi	sannaʔ }.	
kaissaŋaŋ	i	iramaŋ	korupsi	sannaʔ	
known	3ABS	NAME	corrupt	very	

'Ramang is known to be very corrupt '

b.

{$_\iota$ kessaŋa	ĩ	irama**ŋ** },	{$_\iota$ **k**apala	ɹesa },	{$_\iota$ korupsi	sannaʔ } .
kaissaŋaŋ	i	iramaŋ	kapala	desa	korupsi	sannaʔ
known	3ABS	NAME	head	village	corrupt	very

'Ramang, the mayor, is known to be very corrupt '

This result sets up a prediction: if Reinforcer Postposing operates over the surface constituency of the prosody, rather than applying in the syntax and targeting positions that happen to fall at the right edges of the ι, then Reinforcer Postposing should respond to the prosodic effect of appositive insertion—even though this process should have no effect on the clausal syntax. This prediction is correct: when an appositive is inserted after an associate, the associated reinforcer cannot move to the edge of its clause (71a). The grammatical outcome is shown in example (71b): the reinforcer falls at the right edge of the smallest ι that contains its associate—and thus appears before the appositive.

(71) *Appositive Insertion: Disrupts Reinforcer Postposing*

a.

{$_\iota$	*kessaŋa	ĩ	**ɹo**	ramaŋ }, {$_\iota$	kapala	ɹesa }, {$_\iota$	korupsi	sannaʔ	**o**	}.
	kaissaŋaŋ	i	do	ramaŋ	kapala	desa	korupsi	sannaʔ	o	
	known	3ABS	that	NAME	head	village	corrupt	very	there	

INTENDED: 'That Ramang there, the mayor, is known to be very corrupt.'

b.

{$_\iota$	kessaŋa	ĩ	**ɹo**	ramaŋ	**o**	}, {$_\iota$	kapala	ɹesa }, {$_\iota$	korupsi	sannaʔ }.
	kaissaŋaŋ	i	do	ramaŋ	o		kapala	desa	korupsi	sannaʔ
	known	3ABS	that	NAME	there		head	village	corrupt	very

'That Ramang there, the mayor, is known to be very corrupt '

The same effect can be seen in a type of traditional poem called a *Kalinda'da'*. Each *Kalinda'da'* contains four metrical lines and each line forms an independent ι. But it is clear that the prosodic divides between these lines are not linked to any process of syntactic extraposition, as the lines can be freely crossed by WH-dependencies (on a par with their counterparts in English: Thoms 2010). The following examples illustrate in a poem of courtship, where the WH-word *inna* "where" moves between lines 1-2 (72a).

(72) *Mandar Poetry: Intonational Phrase Boundaries without Extraposition*

a.

{$_\iota$	**inna**	wandamo	ɹisaŋa	} {$_\iota$	diposara	watammu	___ }?
	inna	bandamo	di-saŋa		di-posara	bataŋ-mu	
	where	exactly	PASS-think		PASS-take care of	body-2GEN	

'Where exactly do you think that you will be taken care of?'

b.

{$_\iota$	atena	woŋi	} {$_\iota$	mala	di pattindoaŋ	}.
	ate-na	boŋi		mala	di pattindoaŋ	
	heart-3GEN	night		can	in bed	

'In the middle of night, it could happen in bed.' Muthalib & Sangi 1991: 284

As a result, these poems provide a second context where prosodic boundaries fall in positions that have no correspondents in the syntax: metrical line breaks force the emergence of ι boundaries in positions that do not fall at consistent syntactic junctures or correspond to singular syntactic positions. It thus seems impossible to imagine that the reinforcers could be carried to the right edge of a metrical line by any operation of the syntax—and yet in the cases where selecting demonstratives appear in this type of poetry, this is exactly what we find. When a clause like (73a) is parsed into two lines, a reinforcer that originates within the first line is forced to surface at its right edge (73b).

(73) *Reinforcer Postposing interacts with Metrical Line Breaks*

a.

{$_\iota$	jamo	**ɹo**	ɹisaŋa	lopi	pattonda	roppoŋ	**o**	}
	jamo	do	disaŋa	lopi	pattonda	roppoŋ	**o**	}
	is	that	called	boat	shipping	grass	there	

"That there is called a worthless endeavor."

b.

{$_\iota$	jamo	**ɹo**	ɹisaŋa	**o**	}	{$_\iota$	lopi	pattonda	roppoŋ	}
	jamo	do	disaŋa	**o**	}	{$_\iota$	lopi	pattonda	roppoŋ	}
	is	that	called	there			boat	shipping	grass	

"That's called (linebreak) a worthless endeavor." Muthalib & Sangi 1991: 374

The force of these facts is to suggest that there is no way to formulate a syntactic analysis along the lines in (69), which places the reinforcers in syntactic positions that will invariably fall at the right edges of ιs. This is because there is no way to predict or define the positions that fall before appositives and metrical line breaks in the syntax. The understanding that emerges from these facts instead is one on which some reference to surface prosody is absolutely necessary in the formalization of Reinforcer Postposing: in a manner that goes beyond what can be reasonably extrapolated from the syntax, this process must place its targets at the right edge of the ι.

In the face of this result, it is technically possible to formulate a syntactic analysis of Reinforcer Postposing that captures our prosodic generalization through a surface filter: perhaps the syntax carries the reinforcers through a string of many different positions, and the phonology chooses to spell the reinforcers out in whatever link along that chain falls at the edge of the ι (along the lines proposed for certain second-position elements by Bošković 2001). In principle, there is nothing wrong with this type of account—

especially as it is clear that there do exist syntactic operations, like Heavy NP-shift, which are subject to surface filters of this type (Inkelas & Zec, 1995). But within the present investigation, it seems that all of our evidence points in the opposite direction: (*i*) Reinforcer Postposing ignores all types of syntactic islands, evading the CSC, the RRC, and the CSC, and (*ii*) Reinforcer Postposing ignores c-command in the calculus of relative locality, relying instead on linear order alone to determine how reinforcers will move. As a result, I will propose and pursue the hypothesis in (74): one which takes Reinforcer Postposing to operate fully within the phonology and place its targets in a position that is defined purely in terms of the surface prosodic organization of the clause.

(74) *The Phonological Promise*

Reinforcer Postposing is a phonological process that carries reinforcers from their base positions to the right edge of the smallest containing ι.

The immediate implication of this claim is that phonological movement must exist: in other words, the phonology must have the capacity to directly reorder constituents and guide the system of linearization. On some level, this result is not a surprise: the phonology clearly has the capacity to reorder segments (e.g., in cases of metathesis), and at a higher level, it seems to play a decisive role in determining the linear positions of morphemes within the ω (Hargus & Tuttle 1997, and on infixes specifically: McCarthy & Prince 1993) and second-position elements (within the ϕ: Radanović-Kocić 1988, 1996; Halpern 1995; Chung 2003; Harizanov 2014). Even at the level of the intonational phrase, moreover, there is independent reason to believe that the phonology must be able to carry syntactically independent elements across great distances to land at the right edge of the ι—and this is because Aissen 2017 documents a system almost exactly parallel to the one that has unfolded through our study of in Mandar in the Mayan language Tsotsil. In that language, and specifically in the dialect of Zinacantán, Aissen 2017 shows that two definite determiners (and several other elements that contain or resemble them) lexically select an element *e* that always and only appears at the right edge of the ι. It can be seen in that position, for instance, in fragments (75a) and in root clauses (75b).

(75) *Reinforcer Postposing: Tsotsil* (Glosses adapted)

a. $\{_\iota$ *ti* anima j-muk'tot *e* $\}$

DET late 1GEN-grandfather REINFORCER

'My late grandfather' Aissen 2017: 4b

b. $\{_\iota$ Li'abtejotikotik xchi'uk *li* Kumpa Lol ta museo *e* $\}$

we worked with DET compadre NAME at museum REINF.

'We worked with Compadre Lol at the museum.' Laughlin 1980: 25

The distribution of intonational phrases in Tsostil seems to be exactly parallel to the distribution of intonational phrases in Mandar, English, and Indonesian, and as a result, the reinforcers in Tsotsil escape from fronted foci (76a), immediately follow external topics (76b), and stop before constituents that are syntactically extraposed (76c).

(76) *Tsotsil: Movement to the Edge of the Intonational Phrase* (Glosses adapted)

a. $\{_\iota$ Ta sba me *l*-av-ajnil chamuy *e*

on top CLITIC DET-2GEN-wife you should climb REINF.

'It's ON TOP OF YOUR WIFE$_{\text{FOC}}$ that you should climb.' Laughlin 1977: 56

b. $\{_\iota$ *ti* anima j-muk'tot *e* $\}$ $\{_\iota$ x'ok' xa la sutel tal $\}$

DET late 1GEN-grandfather REINF. cry CLITICS returning here

'My late grandfather$_{\text{TOP}}$ returned crying.' Aissen 2017: 21

c. $\{_\iota$ Ikalbe *li* Kumpa Lol un *e* $\}$ $\{_\iota$ ti yu'un chicham xa un e $\}$.

I told DET compadre NAME CLITIC REINF. that I was feeling awful

'I told Compadre Bob that I was feeling awful.' Laughlin 1980: 30

The existence of this pattern, and the existence of seemingly parallel effects across the Mayan family (Skopeteas, 2010), suggests that our rule of Reinforcer Postposing is not unique—and that the phonology must be able to move elements to the edge of the ι.

6 Reinforcer Postposing Revised

To understand how this type of movement in the phonology might actually occur, then, we can return to the theory of triggers and targets. Up to this point, we have treated Reinforcer Postposing as an item-specific and parochial rule—one that targets specific elements and places them in a specific position with no reference to wider systems in the grammar. But now that we have discovered that this process has a consistent phonological shape, we can attempt to leverage the phonology to address two questions:
(*i*) why does Reinforcer Postposing specifically target the right edge of the ι?
(*ii*) why does Reinforcer Postposing specifically target *e/o* and nothing else?

The task of this section is to address these points and develop a purely phonological theory of why Reinforcer Postposing has the exact shape that it does. The investigation that follows, in turn, will gradually lead us to the conclusion that phonological movement can be triggered by the principle of Greed: Reinforcer Postposing occurs to resolve a phonological tension that holds within the moving elements themselves.

6.1 Foot Structure and the Intonational Phrase

We begin with a fact about word-level stress at the right edge of the ι. Mandar is a language that shows regular penultimate stress at the level of the word, and within the native vocabulary, there are few lexical exceptions to this rule (Pelenkahu *et al.*, 1983). But there is a small class of words in the language that follow a different pattern: in ι-medial positions they show regular penultimate stress, but at the right edge of the ι, their stress becomes final. The following examples illustrate with the root *magarriŋ* "have a fever." This root shows penultimate stress when it appears before a subject or any other item in the same ι (77a) but it shows final stress at the right edge of the ι (77b).

(77) *Some Lexical Items: Final Stress at the ι Edge*

a. {$_\iota$ jári i **maɣárrik** kandíʔu }.
jari i magarriŋ kandiʔ-u
DID 3ABS have a fever little sibling-1GEN

'My little sibling DID have a fever.'

b. {$_\iota$ jári i **maɣarríŋ** }.
jari i magarriŋ
DID 3ABS have a fever

'She DID have a fever.'

The same process—the apparent shift to word-final stress at the right edge of the ι—is matched in a much more general way within the system of focus. In the default case, the roots that show regular penultimate stress in Mandar retain their usual stress pattern at the right edge of the ι: thus the verb *cowa* "try" ordinarily surfaces in that position with the stress pattern [t͡ʃówa] (78a). But when roots of this type appear at the right edge of the ι and carry a specific type of contrastive focus, they show the same type of final stress. Under this type of focus, then, the root *cowa* "try" surfaces as [t͡ʃowà] (78a).[4]

(78) *Contrastive Focus: Final Stress at the ι Edge*

a. {$_\iota$ méloʔ i **ut͡ʃówa** }.
meloʔ i u-t͡ʃowa
want 3ABS 1ERG-try

'I want to try it.'

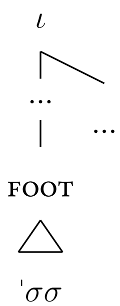
b. {$_\iota$ méloʔ i **ut͡ʃowà** }.
meloʔ i u-t͡ʃowa
want 3ABS 1ERG-try

'I want to TRY it (not BUY it).'

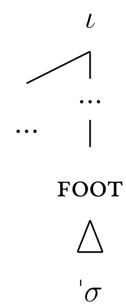
At first blush, it would seem that this pair of effects could be understood in several different ways—given the depth and mystery of the relationships between intonational prominence, focus, and word-level stress (Gussenhoven, 2011; Gordon, 2014). But a second set of patterns in the language open up a direct path to the analysis below: the language prefers syllables to be parsed into disyllabic trochees with the shape in (79a), but at the right edge of the ι, it allows the emergence of monosyllabic feet (79b).

(79) *Right Edge of the Intonational Phrase: Special Foot*

a. *The Default Foot*



b. *The Right Edge of the ι*



The case for this claim—that the right edge of the ι can licence degenerate feet in Mandar—emerges from a system of constraints on minimality. Beneath the level of the ι, there is a smaller prosodic constituent whose right edge can be seen in Mandar from the distribution of a high tone (H). This tone falls at the right edges of many constituents that form XPs in the syntax, and two of these are shown in example (80): the complex NP *itim buku kaiyang* "that big book" and ADV-V sequence like *mane wemme* "just fell."

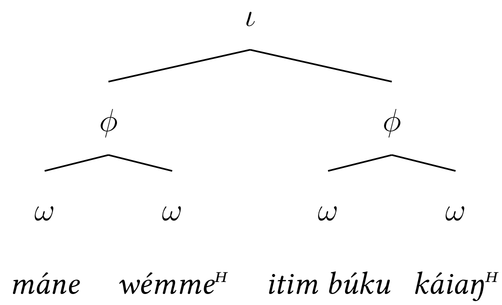
(80) *Another Prosodic Domain*

$\{_\iota$	$\{_{??}$	máne wémme[H]	}	i	$\{_{??}$	itim búku káiaŋ[H]	}	}.
		mane bemme		i		itiŋ buku kaiaŋ		
		just fall		3ABS		that book big		

'That big book just fell.'

Brodkin 2024a,b argues that this H tone falls at the right edge of the phonological phrase (ϕ): the single prosodic constituent that falls between the ι and the ω on the universal prosodic hierarchy Itô & Mester 2007. On this analysis, the distribution of H-tones and word-level stress reveals the prosodic parse in tree (81): the string *mane* *wemme* "just fell" contains two ωs and forms a ϕ, the string *itim buku kaiyang* "that big book" also contains two ωs and forms a ϕ, and both of these ϕs are parsed into an ι.

(81) *Phonological Phrasing in Mandar*



The ϕ is relevant because of a constraint that it imposes at its right edge. Mandar has a large class of functional elements whose underlying forms have the shape (C)V(C), including demonstratives (*de, do*), prepositions, complementizers, and auxiliaries. The elements of this shape canonically surface in monosyllabic form when they appear in the middle of the ϕ, and the following example illustrates this fact with the P^0 *suŋ* "out."

(82) *CVC Heads: Monosyllabic when Medial in the* ϕ

{$_\iota$	{$_\phi$	bémme^{H} }	i	{$_\phi$	**sup**	pepattóaŋ^{H}	}	}.
		bemme	i		suŋ	pepattoaŋ		
		fall	3ABS		out	window		

'It fell out the window.'

When these functional heads are forced to fall at the right edge of the ϕ, however, they are forced to change shape. The essential effect can be seen clearly in the system of complement ellipsis, which can strand many types of functional heads and render them the rightmost overt elements within their corresponding ϕs. The following example shows the ensuing result with the P^0 *suŋ*: when it is stranded by NP-ellipsis, this P^0 is forced to carry the H-tone that would usually fall at the right edge of the PP and it is forced to undergo a process of Vʔ-epenthesis to take on the disyllabic form [*suʔuŋ*].

(83) *CVC Heads: Disyllabic when Final in the ϕ*

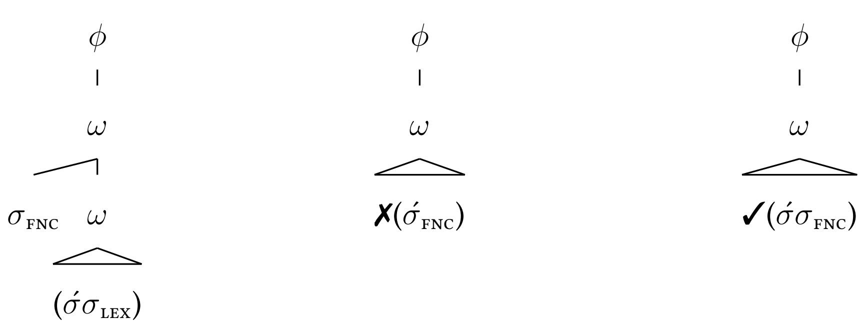
$\{_\iota$	$\{_\phi$	bémme$^{\text{H}}$ }	i	$\{_\phi$	**súʔup**$^{\text{H}}$	___	}	$\{_\phi$	pósa$^{\text{H}}$	}	}.
		bemme	i		suŋ				posa		
		fall	3ABS		out				cat		

'The cat fell out.'

Brodkin 2024b argues that this effect reflects the working of a top-down constraint: the final foot in the ϕ must be a disyllabic trochee. This is a constraint on positional well-formedness (De Lacy, 2001; Smith, 2002), and it has a familiar shape: it requires a low-level prosodic constituent to have a branching structure (the foot must contain two syllables) when it appears at the edge of a higher prosodic domain (the ϕ), on a par with similar effects along the prosodic hierarchy (Booij, 1999; Prieto, 2005; Elordieta, 2006). The following diagram illustrates the key effect. In cases that lack stranding, Brodkin 2024b argues that P^{0}s are parsed into ωs with following N^{0}s in Mandar (as part of a larger pattern at the syntax-prosody interface: Selkirk 1995; Itô & Mester 2009), yielding the parse in (84a). In the context of stranding, ϕs are still built around these PP (again in keeping with a wider pattern: Itô & Mester 2019). In this case, these P^{0}s are forced to form ωs and contain FEET by the architectural requirement of HEADEDNESS (Nespor & Vogel, 1986). At the right edge of the ϕ, the monosyllabic foot in (84b) is banned—and as a result, these heads are forced to be augmented to host the disyllabic foot in (84c).

(84) *The Positional Minimality Constraint*

a. *The Default* b. *The Problem* c. *The Repair*



It is against this backdrop that we return to our shift at the right edge of the ι. When functional heads are stranded at the right edge of the ι and forced to appear at the right edge of a ϕ in that position, they are usually forced to undergo the usual repairs: the head P^0 *suŋ*, for instance, shows Vʔ-epenthesis (85a). But when they are stranded in this position and placed under the type of contrastive focus that descriptively triggers the emergence of word-final prominence in that position, these repairs are suspended—and thus the P^0 *suŋ* retains its usual monosyllabic shape (85b).

(85) *Final Prominence Licenses ϕ-Final Monosyllables*

a.

{$_{\iota}$	{$_{\phi}$	t͡ʃówa	}	i	{$_{\phi}$	sóros$^{\text{H}}$	}	{$_{\phi}$	**súʔuŋ**$^{\text{H}}$ ___$_{\text{NP}}$ } } .
		t͡ʃoba		i		soroŋ			suŋ
		try		3ABS		push			out

‘Try to push it out.’

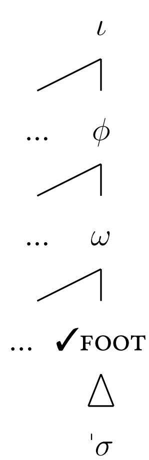
b.

{$_{\iota}$	{$_{\phi}$	t͡ʃówa	}	i	{$_{\phi}$	sóros$^{\text{H}}$	}	{$_{\phi}$	**sùŋ**$^{\text{H}}$ ___$_{\text{NP}}$ } } .
		t͡ʃoba		i		soroŋ			suŋ
		try		3ABS		push			out

‘Try to push it OUT (not IN).’

I propose that the exceptional emergence of ϕ-final monosyllables in this context emerges from a shift in the possible patterns of foot structure: monosyllabic feet are licensed at the right edge of the ϕ only when they also fall at the right edge of the ι.

(86) *Phrase-Final Monosyllabic Feet: OK at the Right Edge of the Intonational Phrase*



6.2 Reinforcer Postposing and Output Optimization

We now return to our study of Reinforcer Postposing with the following results in tow:

(*i*) There is a smaller phonological constituent ϕ beneath the ι, corresponding to the XP, which systematically bans monosyllabic feet at its right edge in Mandar, and

(*ii*) The ban on ϕ-final monosyllabic feet is lifted at the right edge of the ι in two cases:

(*a*) when ϕ-final foci carry an accent that can only appear at the right edge of the ι, or

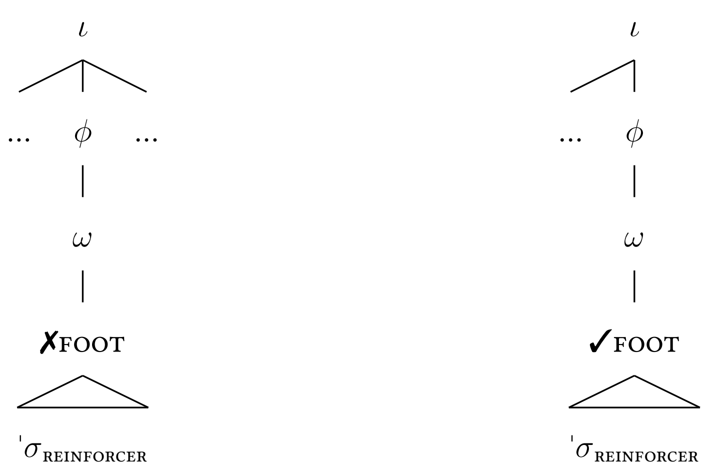
(*b*) when ϕ-final elements are lexically prespecified to host final monosyllabic feet.

With this much in place, we can begin to understand how Reinforcer Postposing might serve as a properly phonological solution to a properly phonological problem. Suppose for a moment that the reinforcers might always form independent ϕs.

If this assumption is correct, then the reinforcers—as monosyllabic elements—should implicitly violate the constraint in (87a): they are monosyllabic, they cannot contain disyllabic feet, and thus they should be banned at the edges of the ϕs that they project. But if they are lexically prespecified to host monosyllabic feet of this type, on a par with roots like *magarring* "have a fever," then we should expect that the reinforcers will be saved from this kind of prosodic problem if they surface at the right edge of the ι (87b).

(87) *Reinforcer Postposing as a Phonological Repair*

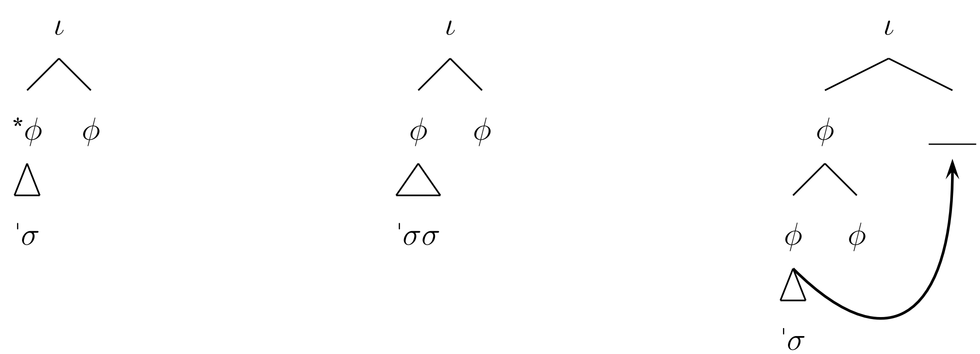
a. *The Problem* b. *The Solution*



The result is that—if it can be shown that the reinforcers always form ϕs—we can understand Reinforcer Postposing to be an operation with an output-optimizing effect: its role is to take elements that are implicitly problematic (ϕ-final monosyllables) and put them in the one position that licenses their particular shape (the right edge of the ι). In other words, this analysis would allow us to understand Reinforcer Postposing as an item-specific repair to a general constraint: the language bans monosyllabic feet at the right edge of the ϕ (88a), requires many heads to be augmented to disyllabic forms in that position (88b), and forces the reinforcers to move to the right edge of the ι instead (88c).

(88) *Reinforcer Postposing: One Repair within a Conspiracy*

a. *Constraint*   b. *General Repair*   c. *Special Repair*



What, then, is the evidence that the reinforcers form their own independent ϕs? The argument is straightforward: the reinforcers carry the pitch contour that falls on every ϕ that appears at the right edge of an ι: a ϕ-final H-tone followed by the boundary tone that falls at the right edge of the ι (in declarative clauses, a simple low tone).

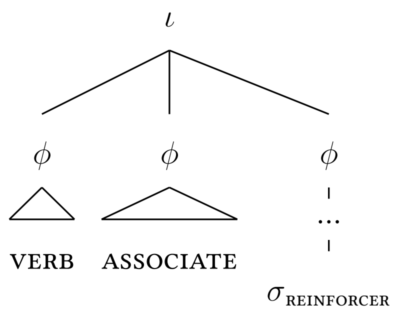
(89) *Tonal Evidence: Reinforcers Form ϕs*

a. $\{_\iota$ $\{_\phi$ lámba$^{\text{H}}$ $\}$ i $\{_\phi$ jáli$^{\text{H-L\%}}$ $\}$ $\}$
go 3ABS NAME
'Ali went.'

b. $\{_\iota$ $\{_\phi$ lámba$^{\text{H}}$ $\}$ i $\{_\phi$ ɹo áli$^{\text{H}}$ $\}$ $\{_\phi$ ó$^{\text{H-L\%}}$ $\}$ $\}$
go 3ABS that NAME there
'That Ali went.'

We can thus be reasonably certain that clauses like (89b) show the rough parse in (90): the reinforcer falls outside the ϕ that corresponds to its associate (even when the reinforcer and the associate are linearly adjacent) and forms an independent ϕ of its own.

(90) *The Prosodic Parse of Reinforcers*



At the syntax-prosody interface, we can understand this parse to emerge directly from the syntactic structure that we have posited up to this point. On the view that the reinforcer is generated in a specifier position within the DP, it follows that it will be both minimal and maximal in the syntax on the theory of Bare Phrase Structure (Chomsky, 1995a). At the syntax-prosody interface, as a result, it should have an ambiguous status with respect to the X^0/XP divide—and should thus be targeted by the constraints that drive the mapping of XPs to ϕs at the syntax-prosody interface. Within the framework of Optimality Theory (Prince & Smolensky, 2004), we can understand this process to proceed in the following way: (*i*) the reinforcers are forced to form ϕs by the Syntax-to-Prosody mapping constraint in (91a), which maps XPs to ϕs (borrowing from Itô & Mester 2019 the intuition that this pressure is one of existential correspondence, rather than alignment; *cf.* Selkirk 2009, 2011; Elfner 2015); (*ii*) the reinforcers are forced to form ωs by both the corresponding Syntax-to-Prosody mapping constraint in (91b) and the constraint Headedness (91c), which demands that prosodic constituents contain heads (Nespor & Vogel, 1986). The ensuing derivation is shown in tableau (91d).

(91) *Mapping: Reinforcer* $\rightarrow \phi$

a. MATCH(XP,ϕ): AOV for every XP in an input syntactic representation S that corresponds to no ϕ in a corresponding output phonological representation P.

b. MATCH(X^0,ω): AOV for every X^0_{LEX} in an input syntactic representation S that corresponds to no ω in a corresponding output phonological representation P.

c. HEADEDNESS: AOV for every nonterminal output prosodic category that does not immediately dominate a prosodic category of equal or immediately lesser size on the prosodic hierarchy $\iota > \phi > \omega >$ FOOT $> \sigma > \mu$.

d. *Tableau: Reinforcers are mapped to* ϕ*s*:

$[_{DEMP}\ [_{DEM'}$ *do Ali* $]\ [_{ADVP}$ *o* $]\ ]$	MATCH(XP,ϕ)	MATCH(X^0,ω)	HEADEDNESS
☞ a. $\{_\iota\ \{_\phi\ [_\omega$ *do (áli)* $]\}\ \{_\phi\ [_\omega$ *(ó)* $]\}\ \}$			
b. $\{_\iota\ \{_\phi\ [_\omega$ do (áli) $]\}\ \{_\phi$ (ó) $\}\ \}$		*!	*
c. $\{_\iota\ \{_\phi\ [_\omega$ do (áli) $]\}$ o $\}$	*!	*	*

We can then formalize the source of the phonological problem that drives our operation: the reinforcers are phonologically illicit outside the right edge of the ι. The first half of this constraint—the ban on ϕ-final monosyllabic feet—emerges from the constraint in (92a), which demands that the ϕ be right-aligned with a disyllabic trochee. The second half of the problem— the resistance of reinforcers to repairs like vʔ-epenthesis—can then emerge from a constraint that mandates the emergence of lexically prespecified metrical structure (Inkelas, 1999), along the lines in (92b)

(92) *The Pressures Behind Postposing*

a. POSITIONAL MINIMALITY (formally ALIGN-R (ϕ,($\acute{\sigma}\sigma$)))

Assign one violation for every ϕ that is not right-aligned with a disyllabic FT.

b. PARSE-FOOT: AOV for every foot in the input absent from the output.

Finally, we can understand the special properties of the right edge of the ι to emerge from a constraint on positional markedness (Zoll, 1996, 1998; Itô & Mester, 1999; Walker, 2001, 2011): one which exacts further violations from ϕ-final monosyllabic feet that appear outside of the right edge of the ι.

(93) *Licensing Monosyllabic Feet at at the Right Edge of the ι*
LICENSE ($(\acute{\sigma})$, ___$\}_\phi\}_\iota$): Outside the right edge of the ι,
assign one violation for every ϕ that is not right-aligned with a disyllabic foot.

We can now see how these constraints deliver the general facts of stress in Mandar. FIRST, restricting our attention to ι-medial contexts, we can guarantee that disyllabic trochees will always be built at the right edge of the ϕ with the ranking of LICENSE > PARSE-FOOT. When the right edge of the ϕ falls after a root that is lexically prespecified to show final stress, like *magarriŋ*, this ranking will force out a disyllabic trochee in that position instead. The following tableau illustrates with the data from example (77a).

(94) *Outside the Right Edge of the ι: ✗Final Stress at the Right Edge of the ϕ*

jari i magar(ríŋ) kandiʔu "My sibling DID have a fever."	LICENSE	PARSE-FOOT
☞ a. $\{_\iota$ $\{_\phi$ $[_\omega$ (jári)]} i $\{_\phi$ $[_\omega$ ma(ɣárrik)]} $\{_\phi$ $[_\omega$ kan(*díʔu*)]} }		$*$
b. $\{_\iota$ $\{_\phi$ $[_\omega$ (jári)]} i $\{_\phi$ $[_\omega$ maɣar(rík)]} $\{_\phi$ $[_\omega$ kan(*díʔu*)]} }	$*!$	

SECOND, turning our attention to the right edge of the ι, we can ensure that lexically prespecified monosyllabic feet will be built at the right edge of the ϕ in this position through the ranking of LICENSE > PARSE-FOOT > POSITIONAL MINIMALITY. The following tableau illustrates, building from the data in example (77b).

(95) *The Right Edge of the ι: ✓Final Stress at the Right Edge of the ϕ*

jari i magar(ríŋ) "She DID have a fever."	LICENSE	PARSE-FOOT	POS-MIN
a. $\{_\iota$ $\{_\phi$ $[_\omega$ (jári)]} i $\{_\phi$ $[_\omega$ ma(ɣárriŋ)]} }		$*$	
☞ b. $\{_\iota$ $\{_\phi$ $[_\omega$ (jári)]} i $\{_\phi$ $[_\omega$ maɣar(ríŋ)]} }			$*$

With this much in place, we can advance on the structure of Reinforcer Postposing. Consider first an example where an associate surfaces in the middle of an ι, like (96a). If its reinforcer immediately follows it in this context, we arrive at the following problem: (*i*) The mapping constraint MATCH(XP,ϕ) forces the reinforcer to map to an ι-medial ϕ, (*ii*) the constraint LICENSE bans the emergence of a monosyllabic foot at its right edge, (*iii*) the reinforcer is lexically prespecified to host a monosyllabic foot, and so (*iv*) the constraint PARSE-FT prohibits the usual repairs to LICENSE (e.g. Vʔ-epenthesis).

(96) *The Problem with ι-Medial Reinforcers*

a. $\{_\iota$ *Tanda i **do** setang **o** dionging $\}$.
arrive 3ABS that demon there yesterday

INTENDED: 'That demon there arrived yesterday.'

b. Tableau: ✗Reinforcers In-Situ

tanda i do setaŋ (ó) dioŋiŋ "That demon arrived yesterday."	MATCH	LICENSE	PARSE
a. $\{_\iota$ … $\{_\phi$ $[_\omega$ do (sétaŋ) $]\}$ $\{_\phi$ $[_\omega$ (ó) $]\}$ … $\}$		$*!$	
☹ b. $\{_\iota$ … $\{_\phi$ $[_\omega$ do (sétaŋ) $]\}$ $\{_\phi$ $[_\omega$ (óʔo) $]\}$ … $\}$			$*!$

For this network of violations to drive the process of Reinforcer Postposing, then, these three constraints must outrank a pressure that bans movement in the phonology—a type of faithfulness constraint that mandates the preservation of the linear ordering relationships established between terminal nodes at the derivational stage where the first round of linearization takes place (plausibly in the morphology (Embick, 2007; Arregi & Nevins, 2012), in the framework of Distributed Morphology (Halle & Marantz, 1993)). I take this constraint to have the shape in (97) (cf. Bennett *et al.* 2016; Kusmer 2020).

(97) *No Shift*
Assign one violation for every pair of terminal nodes in the morphosyntax α, β, such that α is linearized before β in a morphological representation M, for which the output correspondent of α does not precede the output correspondent of β in a corresponding phonological representation P.

The ranking of MATCH(XP,ϕ), LICENSE, and PARSE-FT over NO SHIFT will thus force the process of Reinforcer Postposing: the reinforcers will evade the violations of the first three constraints, which we have already seen, by shifting to the right edge of the ι.

(98) *The Derivation of Reinforcer Postposing*

a. | $\{_\iota$ | Tanda | i | **do** | setang | dionging | **o** | $\}$. |
|---|---|---|---|---|---|---|---|
| | arrive | 3ABS | that | demon | yesterday | there | |

INTENDED: 'That demon there arrived yesterday.'

b. Tableau: Reinforcer Postposing

tanda i do setaŋ (ó) dioŋiŋ	MATCH	LICENSE	PRS	NO SHIFT
a. $\{_\iota$ … $\{_\phi$ $[_\omega$ do (sétaŋ) $]\}$ $\{_\phi$ $[_\omega$ (ó) $]\}$ … $\}$		*!		
b. $\{_\iota$ … $\{_\phi$ $[_\omega$ do (sétaŋ) $]\}$ $\{_\phi$ $[_\omega$ (óʔo) $]\}$ … $\}$			*!	
☞ c. $\{_\iota$ … $\{_\phi$ $[_\omega$ do (sétaŋ) $]\}$ ___ … $\{_\phi$ $[_\omega$ (ó) $]\}\}$				*

The immediate consequence of this analysis—which treats Reinforcer Postposing as a phonological solution to a phonological problem—is that the phonology must have the ability to reorder constituents in response to output-oriented phonological constraints. This conclusion is inconsistent with the intellectual tradition that emerges from the work of Kayne 1994, which restricts the capacity to reorder terminal nodes to the narrow syntax. But is consistent with many separate lines of the theory that reject this view, from the framework of Distributed Morphology, which extends the capacity for (re)linearization to the morphology (Embick & Noyer, 2001), to the many proposals that emerge from work on seemingly phonological and output-optimizing steps of displacement (Hargus & Tuttle, 1997; Halpern, 1995; Anderson, 2000; Bennett *et al.*, 2016; Kusmer, 2020). And given the natural and transparent shape of our phonological analysis of Reinforcer Postposing, it would seem that we can now submit a further argument for the theoretical vision that emerges from that second line of work: movement can occur within the phonology. And in tandem with this result, we can argue, it can be driven in that module of the grammar by the principle of GREED.

7 Locality and Ellipsis

With slightly more architectural scaffolding around this account, we can now address the questions that remain. The core of our analysis is the claim that Reinforcer Postposing emerges from top-down interactions between constraints on the structure of feet: the reinforcers are lexically specified to form monosyllabic feet, they are invariably mapped to ϕs, and they shift to the right edge of the ι to retain their monosyllabic shape. The result is that the process of Reinforcer Postposing must apply at a derivational stage that has access to top-down prosodic structure and specifically the distribution of ιs—prosodic constituents that routinely span matrix and embedded clauses. The necessary move, then, is to situate this process within a fully global phonological evaluation that operates over the entire output of the morphosyntactic derivation.

I assume that the cycle does not exist and that all phonological evaluation proceeds in parallel within this single round of global evaluation. My reasons for this are twofold. First, there is no evidence for the existence of a cycle in the realm of phrasal phonology (Kiparsky, 1985) and no empirical benefit to postulating its existence: for instance, there is no way to derive the distribution of ϕs from putative patterns of phase-based spell-out (Cheng & Downing, 2016). Second, many aspects of phonological evaluation operate in an irreducibly parallel way even at the level of the word (McCarthy, 2011; Wei & Walker, 2020; Adler & Zymet, 2021)—and many word-level processes interact with top-down prosodic structure in ways that require lookahead on theories that posit a cycle (Prince, 1975; Dresher, 1983; Selkirk, 1995; Kenstowicz, 2005; Henderson, 2012; Royer, 2022). As a result, I assume with Prince & Smolensky 2004 and the main theoretical current beneath Match Theory (Selkirk, 2009, 2011; Elfner, 2012, 2015) that the construction of prosodic structure and all other tasks of the phonology proceed in a single global and parallel evaluation. I assume further that it is at this stage that Reinforcer Postposing applies.

7.1 Constraints on Locality

Within this single round of phonological evaluation, I assume that output-oriented constraints are evaluated over phonological representations that retain no information about the morphosyntax beneath them (the principle of SYNTAX-FREE PHONOLOGY: Nespor & Vogel 1986). The consequence is that phonological constraints like NO SHIFT, which compare input morphosyntactic representations to output phonological representations, will not be able to detect whether or not reinforcers have crossed over the edges of syntactic islands as they evaluate their positions in surface phonological representations. In the cases where the reinforcers cross out of islands, then, constraints like NO SHIFT will not be able to impose any special penalty. The essential situation is sketched below: in a case where Reinforcer Postposing descriptively violates the CSC, like (99a), NO SHIFT will assign a single violation for each terminal node crossed by that operation—on a par with steps of movement that cross no island boundaries at all (99b).

(99) *Analysis: How Reinforcer Postposing ignores Syntactic Islands*

a. {$_\iota$ U-ita i [$_{\&P}$ sola-na iKaco' na **do** tau] dionging **o** }.
1ERG-see 3ABS friend-3GEN NAME and that guy yesterday there
'I saw Kaco's friend and that guy there yesterday.'

b. Tableau: ✗Syntactic Islands

uita i solana ikat͡ʃoʔ na do tau (ó) dioŋiŋ	OTHER CONSTRAINTS	NO SHIFT
a. {$_\iota$ wita i solana ikat͡ʃoʔ na ɹo tau o djoŋiŋ }	*!	
☞ b. {$_\iota$ wita i solana ikat͡ʃoʔ na ɹo tau ___ djoŋiŋ o }		*

This formalization of NO SHIFT will also deliver the observation that Reinforcer Postposing operates within islands of its own: it always places its targets at the right edge of the smallest possible ι. In clauses where associates are topicalized, then, it carries the reinforcers to the right edge of the ι that corresponds to the topic and proceeds no farther than that (100a). The tableau in (100b) shows how No Shift delivers this second effect: however many violations it assigns for movement to the right edge of the smallest ι, it will always assign MORE violations for movement to the right edge of any other ι (even if we assume, as seems reasonable, a recursive-ι parse for clauses like (100a)).

(100) *Analysis: How Reinforcer Postposing obeys Phonological Islands*

a. {$_\iota$ {$_\iota$ **Do** sanaeke **o** }, {$_\iota$ biasa i m-angino ___ } }.
that kid there usually 3ABS ANTIP-play
'That kid there$_{\text{TOP}}$ is usually playing'

b. Tableau: ✓Phonological Islands

do sanaeke (ó)$_{TOP}$ biasa i maŋino	NO SHIFT
a. {$_\iota$ {$_\iota$ do sanaeke ___ } {$_\iota$ bjasa i maŋino o } }	*!
☞ b. {$_\iota$ {$_\iota$ do sanaeke o } {$_\iota$ bjasa i maŋino } }	

We can leverage the same logic to capture the attested pattern of Relative Locality. In the clauses that contain multiple associates within a single ι, we have seen, it is the rightmost reinforcer that shifts to the edge (101a). Momentarily setting the fate of the other reinforcer aside, we can derive this pattern from the logic above: No Shift assigns one violation for each terminal node crossed by Reinforcer Postposing, and as a result, it will always favor derivations where the rightmost reinforcer moves (101b).

(101) *Analysis: How Reinforcer Postposing ignores C-Command*

a. {$_\iota$ Bemme i **do** sanaeke naung **ndi** passauang **e** }!
fall 3ABS that kid down this well here
Intended: 'That kid there fell down this well here!'

b. Tableau: Purely Linear Relative Locality

bemme i do sanaeke (ó) nauŋ ndi passauaŋ (é)	NO SHIFT
☞ a. {$_\iota$ bemme i do saneke ___ non ndi passauaŋ e }	
b. {$_\iota$ bemme i do saneke ___ non ndi passauaŋ ___ o }	*!**

7.2 Phonological Ellipsis

Turning now to the matter of deletion, we can begin from the generalization in (102).

(102) *Reinforcer Deletion: The Final Statement*
If a reinforcer cannot be realized at the right edge of the ι, it is suppressed.

In the OT analysis developed here, this generalization reduces to the following claim: when the prespecified metrical structure of a reinforcer cannot be licensed by movement to the edge of the ι, the phonology favors deletion of the reinforcer over any other repair.

To understand this effect, we will first have to do some phonological footwork. Back in Tsotsil, Aissen 2017 demonstrates that the morphemes that behave like the Mandar reinforcers —originating in the extended NP and moving to the right edge of the ι—show the same pattern of deletion. In the clauses that contain multiple reinforcers in Tsotsil, Aissen 2017 shows that only one reinforcer surfaces at the right edge of the ι (103).

(103) *Reinforcers in Tsotsil: Suppressed Outside the Right Edge of the ι* (Glosses adapted)

{$_\iota$	Sjipan	la	taora	*ti*	ok'il	__	*ti*	t'ul	__	un	___	*e* }.
	tied	CLITIC	right away	DET	coyote		DET	rabbit		CLITIC		REINFORCER

'The rabbit tied Coyote up right away.' Laughlin 1977:160

In Tsotsil, Aissen 2017 proposes that this pattern emerges from a phonological process of coalescence: multiple reinforcers may move to the right edge of the ι, but a string of reinforcers that are all shaped [*e*] will always coalesce into a single segment in the surface phonology. The result is a second way to understand our pattern of deletion: rather than assuming that reinforcers delete when they are non-final in the ι, we might imagine that they are simply rendered invisible by the regular segmental phonology.

To extend this type of analysis to Mandar, then, we would have to defend two claims. (*i*) in the cases where multiple segmentally identical reinforcers originate in a single ι, they are reduced by a type of coalescence that collapses them together (104a), and (*ii*) in the cases where multiple segmentally distinct reinforcers originate in a single ι, they are reduced by coalescence of sequences like /*eo*/ and /*oe*/ to [*e*] (104b) or [*o*] (104c).

(104) *Mandar Reinforcers: Phonological Coalescence?*

a. $\{_\iota$ nasaka i **ɹe** posa ___ **ɹe** walao ___ **e** }.
nasaka i de posa de balao **ee**
3ERG-catch 3ABS this cat this mouse

'This cat caught this mouse.'

b. $\{_\iota$ bemme i ɹo saneke ___ non ndi passauaŋ ___ **e** }!
bemme i do sanaeke nauŋ ndi passauaŋ **oe**
fall 3ABS that kid down this well
'That kid there fell down this well here!'

c. $\{_\iota$ woloi i **de** potona solana **ɹo** tau ___ ___ **o** }.
u-oloi i de poto-na sola-na do tau **eo**
1ERG-like 3ABS this photo-3GEN friend-3GEN that person

'I like this photo of the friend of that guy there.'

In Mandar, I believe, there is good reason to think that this analysis is not correct. The case emerges from two facts. First, there are no patterns of reduction that operate in the segmental phonology and reduce the sequences /*eo*/ and /*oe*/. The following examples illustrate the basic issue. In the middle of the ϕ, there is an active process of coalescence that reduces the sequences /*ao*/ and /*ae*/ to [*o*] or [*a*] respectively (105a), and this process and its interaction with prosodic phrasing are described in Brodkin 2024a. In the same position, however, the sequences /*eo*/ and /*oe*/ cannot reduce (105b)-(105c).

(105) *Against Coalescence: No Such Process*

a. $\{_\phi$ bal**o** kaiaŋ }, $\{_\phi$ san**e**ke maliŋgao }
bal**ao** kaiaŋ san**ae**ke maliŋgao
mouse big kid tall

'Big mouse, tall kid'

b. {$_\phi$ burum mal**eo** kaiaŋ }, {$_\phi$ bitt**oe**k kett͡ʃuʔ }
burun maleo kaiaŋ bitt**oe**ŋ kett͡ʃuʔ
bird NAME big star small

'Big maleo bird, small stars'

c. {$_\phi$ utum**e** womi }, {$_\phi$ naw**eo**r bomi }, {$_\phi$ miŋŋ**oe** womi }
u-tum**ae** bomi na-b**eo**r bomi miŋ-ŋ**oe** bomi
1ERG-propose again 3ERG-shift again VBLZ-yawn again

'I proposed to her again, he shifted it again, she yawned again'

Second, there is an explicit ban on coalescence of the reinforcers with preceding lexical material. The following examples illustrate. It is generally impossible in Mandar for strings of adjacent identical vowels to surface in hiatus (Brodkin, 2024b), and as a result, it seems reasonable to imagine that sequences like [*oo*] might be forced to coalesce by the segmental phonology. But in the cases where reinforcers should be expected to surface after constituents that end in segmentally identical vowels, coalescence is not what we see. Instead, when the reinforcer *o* should surface after a ω that ends in [*o*], the segment [ʔ] is epenthesized between the two (106a). The same process also operates in the clauses where *e* surfaces after a ω that ends in [*e*] (106b).

(106) *Against Coalescence: Hiatus Resolved by Epenthesis*

a. {$_\iota$ nasaka i **ɹe** posa **ɹo** walao**ʔ** **o** }.
na-saka i de posa do balao **o**
3ERG-catch 3ABS that cat that mouse there

'That cat caught that mouse.'

b. {$_\iota$ nawokko i **ɹo** balao **ɹe** talaɣae**ʔ** **e** }.
na-bokkoʔ i do balao de talagae **e**
3ERG-bit 3ABS that mouse this tomato here

'That mouse bit this tomato here.'

I take these facts to reaffirm our original characterization of the pattern in Mandar: when multiple reinforcers originate within a single ι, they do not pile up and coalesce. Instead, the reinforcers that cannot surface at the right edge of the ι are forced to delete.

With this conclusion in place, we can now go on to observe that the same pattern—"surface at the right edge of the ι or delete"—is neatly attested elsewhere in the language. Mandar has a set of verbs that surface in their transitive forms with a transitive suffix *-i*. The following example illustrates the distribution of this affix with the root *pangino* "play", whose intransitive form we have already seen in examples like (100a).

(107) m-angino, na-pangino-i
ANTIP-play 3ERG-play-TRANSITIVE
'Play (intransitive) / play (transitive)

The transitivity suffix *-i* is useful to our investigation for two interlocking reasons. First, when this suffix appears at the right edge of the ι, it invariably hosts the type of final stress that we have seen to emerge in that position in roots like *magarriŋ*.

(108) *The Transitivity Suffix -ì: Final Stress at the Right Edge of the* ι

a. {$_\iota$ {$_\phi$ bjasa$^{\text{H}}$ } i {$_\phi$ napaŋinoì$^{\text{H}}$ } }.
biasa i na-paŋino-i
usually 3ABS 3ERG-play-TRANSITIVE
'They usually play it.'

b. {$_\iota$ {$_\phi$ mwaʔ itin$^{\text{H}}$ } {$_\phi$ napaŋinoì$^{\text{H}}$ } }, {$_\iota$ {$_\phi$ mitt͡ʃoe$^{\text{H}}$ } aʔ }.
muaʔ itiŋ na-paŋino-i miŋ-t͡ʃoeʔ aʔ
if that 3ERG-play-TRANSITIVE ANTIP-follow 1ABS
'If that's what they're playing, I'm coming.'

Second, whenever this suffix cannot appear at the right edge of the ι—for instance, when the verb is followed by a non-extraposed adjunct or CP—this suffix must delete.

(109) *The Transitivity Suffix -ì: Deletes outside the Right Edge of the* ι

a. {$_\iota$ {$_\phi$ bjasa$^{\text{H}}$ } i {$_\phi$ napaŋino$^{\text{H}}$___ } {$_\phi$ dio$^{\text{H}}$ } }.
biasa i na-paŋino-i dio
usually 3ABS 3ERG-play-TRANSITIVE there
'They usually play it there.'

b.	$\{_\iota$	$\{_\phi$	mala$^{\text{H}}$ }	i	$\{_\phi$	mupaŋino$^{\text{H}}$___	} $\{_\phi$	mwaʔ	dio$^{\text{H}}$ }	aʔ	}.
			mala	i		mu-pangino-i		muaʔ	dio	aʔ	
			can	3ABS		2ERG-play-TRANSITIVE		if	there	1ABS	

‘You can play it when I’m there.’

Exactly the same pattern is attested in many languages of the Mayan family, where certain verbal suffixes that mark transitivity are deleted outside the right edge of the ι (Henderson, 2012; Royer, 2022). Henderson 2012, for instance, presents the following paradigm in K’ichee’: many transitive verbs surface with the transitive “status suffix” *-o* in the contexts where this suffix can be realized at the right edge of the ι (**??**). But in the cases where this suffix cannot be realized at the right edge of the ι, it is suppressed (**??**).

(110) *Parallel Pattern: “Status Suffixes” in K’ichee’* (Glosses adapted)

a.	$\{_\iota$	Xutij-**o**	}.
		he ate-TRANSITIVE	

‘He ate it.’

b.	$\{_\iota$	Xutij-___	le	wah	}.
		he ate-TRANSITIVE	the	tortilla	

‘He ate the tortilla.’

We thus return to our reinforcers with the understanding that they are not unique: there are many cases where morphemes underlyingly present in the morphosyntax are suppressed at the surface when things in the phonology go wrong. Following Kurisu 2001; Raffelsiefen 1996, 1999, 2004, and Henderson 2012, we can understand these effects to emerge from the interaction of output-oriented constraints of the phonology and a violable requirement which demands that terminal nodes be exponed. In other words, we can treat these patterns of suppression as a phonological sort of Rescue-by-Deletion (Ross, 1969; Chomsky, 1972; Merchant, 2001; Bošković, 2011): phonological problems would arise if these elements were realized outside the right edge of the ι, and to rescue the derivation from the ensuing crash, the phonology deletes these elements instead.

What, then, is the violable constraint that forces terminal nodes to exponed? With slightly more architectural footwork, we can understand this pressure in a new way. Bennett *et al.* 2019 propose, in a study of ellipsis in Irish, that the calculus of ellipsis is handled at the interface between the morphosyntax and the phonology. Following the tradition of work that takes ellipsis sites to contain all the usual morphosyntactic structure (Williams, 1977; Fiengo & May, 1994; Merchant, 2001), they propose that the process of ellipsis is implemented in the phonology—where it is forced by a constraint that bans the realization of material that has been marked for ellipsis in the syntax (111a). But if the phonology is indeed responsible for ellipsis, we must also imagine that this constraint is counterbalanced by a second pressure that prevents the gratuitous ellipsis of material that has not been marked in the same way, along the lines in (111b)

(111) *The Constraints that Regulate Ellipsis*

a. ELIDE($X_{[\emptyset]}$): Assign one violation mark for each [$\emptyset$]-marked node that is realized with phonological content in the output. (Bennett *et al.* 2019:58)

b. *ELIDE: Assign one violation for every node not realized in the phonology.

Against this backdrop, I propose the following analysis of our pattern of suppression. Starting with the suffix *-i*, I propose that this head is lexically prespecified to form a monosyllabic foot. When it would appear outside the right edge of the ι, then, it runs into trouble with LICENSE and PARSE. I thus propose that these constraints force a consequent violation of *ELIDE—yielding a type of ellipsis forced by the phonology.

(112) *Reinforcer Deletion: Phonological Ellipsis*

biasa i napaŋino-(í) dio	LICENSE	PARSE	*ELIDE
☞ a. {$_\iota$ bjasa i {$_\phi$ napaŋino-___ } dío }			*
b. {$_\iota$ bjasa i {$_\phi$ napaŋi(nói) } dío }		*	
c. {$_\iota$ bjasa i {$_\phi$ napaŋino(í) } dío }	*!		

Turning now to the reinforcers, we can now capture the pattern of deletion with the same account. In the contexts where reinforcers cannot surface at the right edge of the ι, they too run into trouble with LICENSE and PARSE. We can thus derive the suppression of non-final reinforcers from the same ranking of these constraints above *ELIDE.

(113) *Reinforcer Deletion: Phonological Ellipsis*

bemme i do sanaeke (ó) nauŋ ndi passauaŋ (é)	LICENSE	PARSE	*ELIDE
☞ a. {$_\iota$ bemme i do sanaeke ___ non ndi passauaŋ e }			*
b. {$_\iota$ bemme i do sanaeke {$_\phi$ (oʔo) } non ndi passauaŋ e }		*!	
c. {$_\iota$ bemme i do sanaeke {$_\phi$ (o) } non ndi passauaŋ e }	*!		

7.3 The Interaction with Rightward Focus Shift

This analytical step allows us to resolve the final mystery that remains: the fact that the reinforcers vanish in the domains where other elements undergo Rightward Focus Shift. The solution is now straightforward. There are two surface requirements imposed upon the constituents that undergo Rightward Focus Shift: (*i*) they must be final in the ι and (*ii*) they must carry the type of final stress that falls on contrastive foci in that position.

(114) *Rightward Focus Movement → Final Accent*

{$_\iota$	uweŋa	ĩ	___	saneke	poloppèŋ}.
	u-beŋaŋ	i		sanaeke	poloppeŋ
	1ERG-give	3ABS		kid	pen

'I'm giving the kids PENS.'

This pair of requirements cannot be met in ιs that contain overt reinforcers. This is because overt reinforcers are subjected to a very similar requirement: they must surface at the right edge of the ι. We thus have a simple and surface-oriented explanation for the ungrammaticality of examples like (115): the reinforcers and the targets of Rightward Focus Shift "compete" to surface at the right edge of the ι and they cannot both win.

(115) *Rightward Focus Movement: Incompatible with Overt Reinforcers*

{$_\iota$	*naweŋa	ĩ	ɹo	guru	___	saneke	poloppèŋ	**ò**	}}.
	na-beŋaŋ	i	do	guru		sanaeke	poloppeŋ	o	
	3ERG-give	3ABS	that	teacher		kid	pen	there	

INTENDED: 'That guy there gave the kids PENS.'

We can understand this set of restrictions to emerge from the following constraints. The first is a constraint that requires this particular ι-level accent to fall on the targets of Rightward Focus Shift (116a)—a constraint that forms part of a broader family, familiar from work on Germanic (Truckenbrodt, 1995), which requires various types of foci to be aligned with high-level prosodic events. We can then restrict the distribution of this accent to the right edge of the ι with the run-of-the-mill alignment constraint in (116b). The combined force of these constraints, shown in tableau (116c), is force the targets of Rightward Focus Shift to surface at the absolute right edge of the ι. Ranked above the constraint *ELIDE, in turn, they will force deletion of the reinforcers in clauses like (115).

(116) *Deriving the Complementarity Between Rightward Focus Shift and Overt Reinforcers*

a. ALIGN-FOCUS-ACCENT
Assign one violation for every contrastive focus that lacks this ι-level accent.

b. ALIGN-RIGHT-ACCENT
Assign one violation for every instance of this accent that is not final in the ι.

c. *Rightward Focus Shift forces Reinforcer Deletion*

naweŋaŋ i do guru o ___ sanaeke poloppeŋ	AL-FOC-ACC	AL-R-ACC	*ELIDE
☞ a. {$_\iota$ naweŋa ĩ ɹo ɣuru __ saneke poloppèŋ }			∗
b. {$_\iota$ naweŋa ĩ ɹo ɣuru __ saneke poloppèŋ ò }		∗!	
c. {$_\iota$ naweŋa ĩ ɹo ɣuru __ saneke poloppeŋ ò }	∗!		

The result is an analysis that captures the apparent complementary distribution between Reinforcer Postposing and Rightward Focus Shift without positing a link in the syntax: the two compete in an output-oriented calculus that forces reinforcers to delete.

8 Conclusion

With these results in place, we can now begin to integrate this overall picture with the guiding questions that we staked out in Section 4. Focusing first on these three targets:

1. Provide a unified and exhaustive characterization of the domains of placement, which separates the constituents that host reinforcers from those which do not,
2. Explain how this movement can skirt all syntactic constraints on locality, and
3. Explain why this movement obeys the idiosyncratic constraints that it does.

We can say that (*i*) the reinforcers move to the right edges of their intonational phrases, (*ii*) they evade syntactic locality constraints because they move in the phonology, and (*iii*) they show a purely linear form of relative locality and are trapped within their own intonational phrases because the sole constraint that holds over Reinforcer Postposing is that it implicate the minimum possible amount of relinearization within the phonology.

Turning next to the second group of desiderata,

4. Connect the motivation for postposing to independent properties of the grammar,
5. Capture the Sour-Grapes effect, explaining why reinforcers delete if non-final, &
6. Explain the complementarity of Reinforcer Postposing & Rightward F-Movement.

We can say that (*iv*) The reinforcers are lexically specified to form monosyllabic feet, but they are mapped to ϕs and violate a constraint against ϕ-final monosyllabic feet. They are thus forced to move to the position that licenses their shape—the right edge of the ι. (*v*) Whenever they cannot reach that position, they are suppressed in the phonology in response to output-oriented constraints, in a type of phonological rescue-by-deletion, and (*vi*) in the context of Rightward Focus Movement, they are prevented from reaching their target position by the output constraints that govern the targets of that operation.

Finally, then, we come to the final targets of our investigation:

7. To situate our analysis within a general theory of displacement in the grammar

8. and to bring its results to bear on the theory of the ultimate motivation for Move.

Starting with (*vii*), we can state that the principle result of our investigation is to deliver a novel argument that movement can operate in the phonological component.

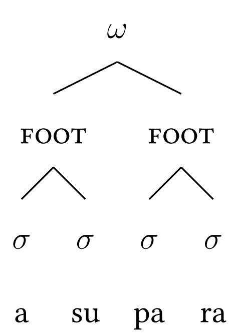
On an empirical level, we have observed, this discovery dovetails with the results of a wide literature on phonological displacement (McCarthy & Prince, 1993; Halpern, 1995; Hargus & Tuttle, 1997; Chung, 2003; Harizanov, 2014; Bennett *et al.*, 2016; Aissen, 2017; Kusmer, 2020) and lends strength to the theoretical vision that emerges from that work.

On a theoretical level, however, I submit that we should also understand this conclusion to follow from and reconfirm the Minimalist hope that operations of the syntax—like movement—might emerge from general mechanisms distributed across the grammar.

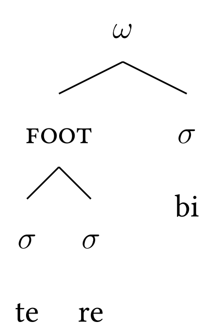
In the realm of prosody, for instance, it is absolutely clear that the phonology must make use of operations like Merge and Adjoin to build the hierarchical prosodic structure that we have studied above—and to deliver the well-known patterns of balanced and imbalanced sisterhood within that domain (Itô & Mester, 1992, 2007; Selkirk, 2009). For example, the phonology must be able to both merge feet with each other (117a) and adjoin syllables to feet (117b) to deliver the productive patterns of truncation in Japanese.

(117) *Symmetric and Asymmetric Structure-Building* (Itô & Mester, 1992)

a. *Asparagus* (/asuparagasu/)



b. *Television* (/terebizjoɴ/)



On Chomsky 2004's view that movement reflects a subcase of Merge and and should thus "come free" in any system that already has Merge, then, we should expect—and now we can confirm—that the same possibility should be extended to the phonology.

Turning next to (*viii*), we can draw our investigation to a close with one final result. In Section 6, we saw that Reinforcer Postposing must be driven by an implicit property of the reinforcers themselves—those elements carry within them a tension that forces them to move or delete. This property sets our process apart from the many cases of phonological movement that have been argued to be driven by push factors, and in particular, the need for prosodically weak elements to escape prosodically prominent positions (Halpern, 1995; Harizanov, 2014; Bennett *et al.*, 2016; Kusmer, 2020). But it establishes a pair of direct parallels with a second class of movement operations: those types of movement that operate in the syntax in response to the principle of Greed. The most important class of operations that have been argued to show this property are the head-movements that proceed by substitution (Rizzi & Roberts, 1989; Ackema *et al.*, 1993; Fanselow, 2004, 2009; Surányi, 2005, 2007; Harizanov & Gribanova, 2019), like the process of T-TO-C in Germanic. In different ways, Fanselow 2004, 2009 and Surányi 2005, 2007 argue that this type of movement must be driven by features on the moving head, rather than any particular attracting properties of the target position. Taking the same view, Lasnik 1999 observes that T-TO-C movement shows a second property in English that is very similar to what we have seen in our study of the Mandar reinforcers: there are contexts where the need for T^0 to move to C^0 is apparently resolved by deleting T^0. Lasnik 1999 argues that this second path to resolution is what lies behind the fact in (118): in the context of matrix sluicing, which Lasnik 1999 takes to involve ellipsis of the TP, auxiliaries do not survive—and thus, on his view, cannot possibly have moved to C^0.

(118) a. Mary will see someone.

b. $[_{CP}$ who $[_{C'}$ __ $_{C^0}$ $[_{\cancel{TP}}$ ~~*will*~~ $_{T^0}$ ~~Mary see~~]]]?

The force of these facts is to suggest that movement must be able to operate in an essentially greedy way in the syntax. In the minimalist way that we should expect to see cross-modular parallelism in this domain, then, it is natural to imagine that movement must be able to operate out of Greed in the phonology as well. It is thus a heartening result of our investigation to see that this final architectural prediction is borne out.

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Notes

[1] GLOSSING: ABS: absolutive, ANTIP: antipassive, ERG: ergative, GEN: genitive. <c> = /t͡ʃ/, <’> = /ʔ/.

[2] There is a second line of work that places demonstratives in specifier positions within the DP (Brugè *et al.*, 1996; Brugè, 2002; Kayne, 2005); analyses of demonstrative-reinforcer constructions in this tradition typically posit structures different from the above (Bernstein, 1997; Roehrs, 2010; Leu, 2015). The case for my analysis rests on a correspondence between syntax and phonology: (*i*) the relevant demonstratives in Mandar are parsed into prosodic words with following nouns, just as other functional X^0s are parsed into prosodic words with their complements in Mandar and beyond, and (*ii*) unlike demonstratives, DP-internal specifiers are generally parsed into independent phonological phrases in Mandar. To the best of my knowledge, however, nothing below will depend on this particular analysis of demonstratives.

Turning to the disyllabic adverbs, there are many ways in which we might understand their positions within the DP. Because they are optional, I will assume that they occupy adjunct positions, unlike the selected reinforcers in (18). Nothing in the following analysis depends on this assumption, however, and many further additional analyses could also be considered (building from, e.g., Bernstein 1997; Roehrs 2010.

[3]The stranding analysis would have to stipulate that CP-preposing cannot strand Reinforcers, but this would undermine its only plausible analysis of violations of the RRC (reinforcer-stranding leftward movement of CPs). The remnant movement analysis, in turn, would be forced to posit that leftward-shifted reinforcers must pull up their associated CPs, restating a generalization on underlying constituency in linear terms.

[4]Space constraints prevent a full description of the phonology of focus in Mandar, but the key generalizations are these: (*i*) this specific accent is restricted to the right edge of the ι; (*ii*) contrastive foci can appear outside of the ι-final position, and in that context the phonological diagnostics for phonological phrasing described in Brodkin 2024a reveal the following: (*a*) maximal phonological phrase boundaries are inserted after contrastive foci (focus-by-alignment (Féry, 2013)), (*b*) all expected maximal phonological phrase boundaries in the space before contrastive foci are erased, and (*c*) all expected maximal phonological phrase boundaries in the space after contrastive foci remain. The fine phonetics of focus are unknown to me, and it is likely that different subtypes of focus have different phonological and phonetic signatures.